1. Objective

This report documents the design and implementation of a multi-source, enterprise-level Business Intelligence system using Tableau Desktop. The project simulates real-world data modelling, advanced analytics, predictive forecasting, cohort analysis, customer segmentation, and performance optimization using the Europe Sales Records dataset.

Dataset Overview

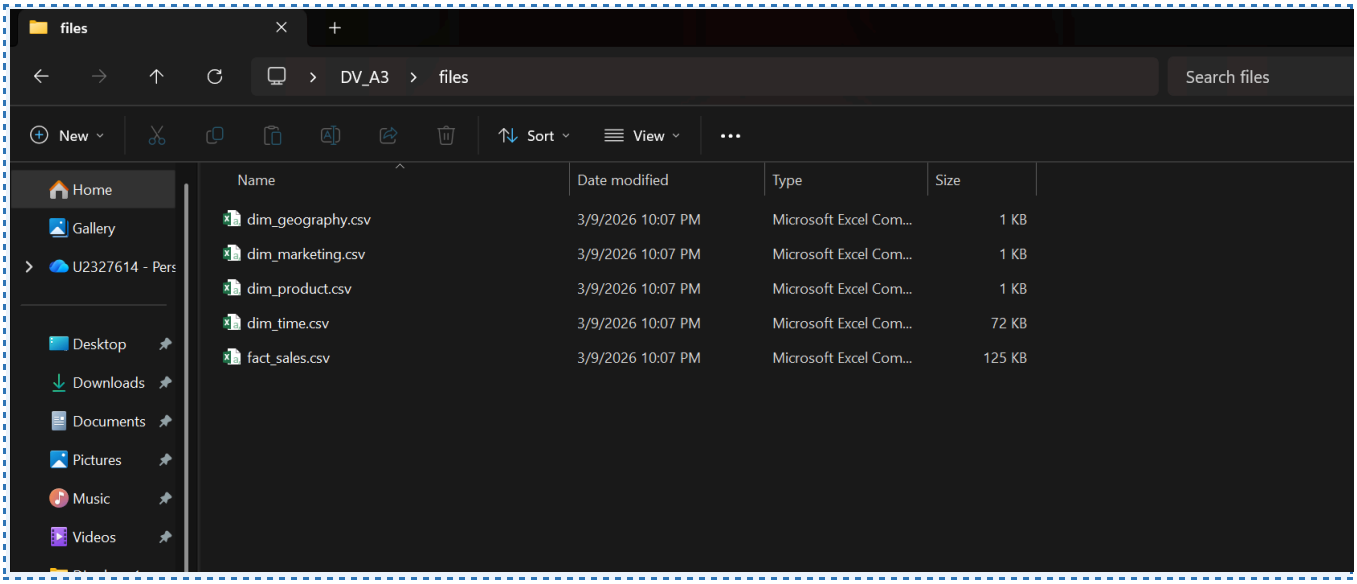
File	Europe_Sales_Records.csv
Total Records	1,330 transactions
Region	Europe (48 countries)
Time Period	2010 – 2017
Key Measures	Total Revenue, Total Cost, Total Profit, Units Sold
Dimensions	Country, Item Type, Sales Channel, Order Priority

Key Assumptions Made

- No Customer ID column existed in the source data. Country was used as the customer proxy unit for all customer-level analytics (cohort, RFM, CLV).
- Order ID was treated as a unique transaction identifier. Surrogate keys were generated as SK-0001, SK-0002, etc.
- The reference date for Recency calculation was set to January 1, 2015 (approximate dataset end).
- Since no negative profit values existed, Profit Leakage was redefined as low profit margin (below 10% = At Risk, 10-20% = Low Margin).
- SCD Type 2 was simulated conceptually using Order Date and Ship Date as versioning fields.

2. Dataset Architecture — Star Schema

A Star Schema was designed by decomposing the original flat CSV into one Fact Table and four Dimension Tables. Each dimension table was generated programmatically and linked to the fact table via foreign keys.



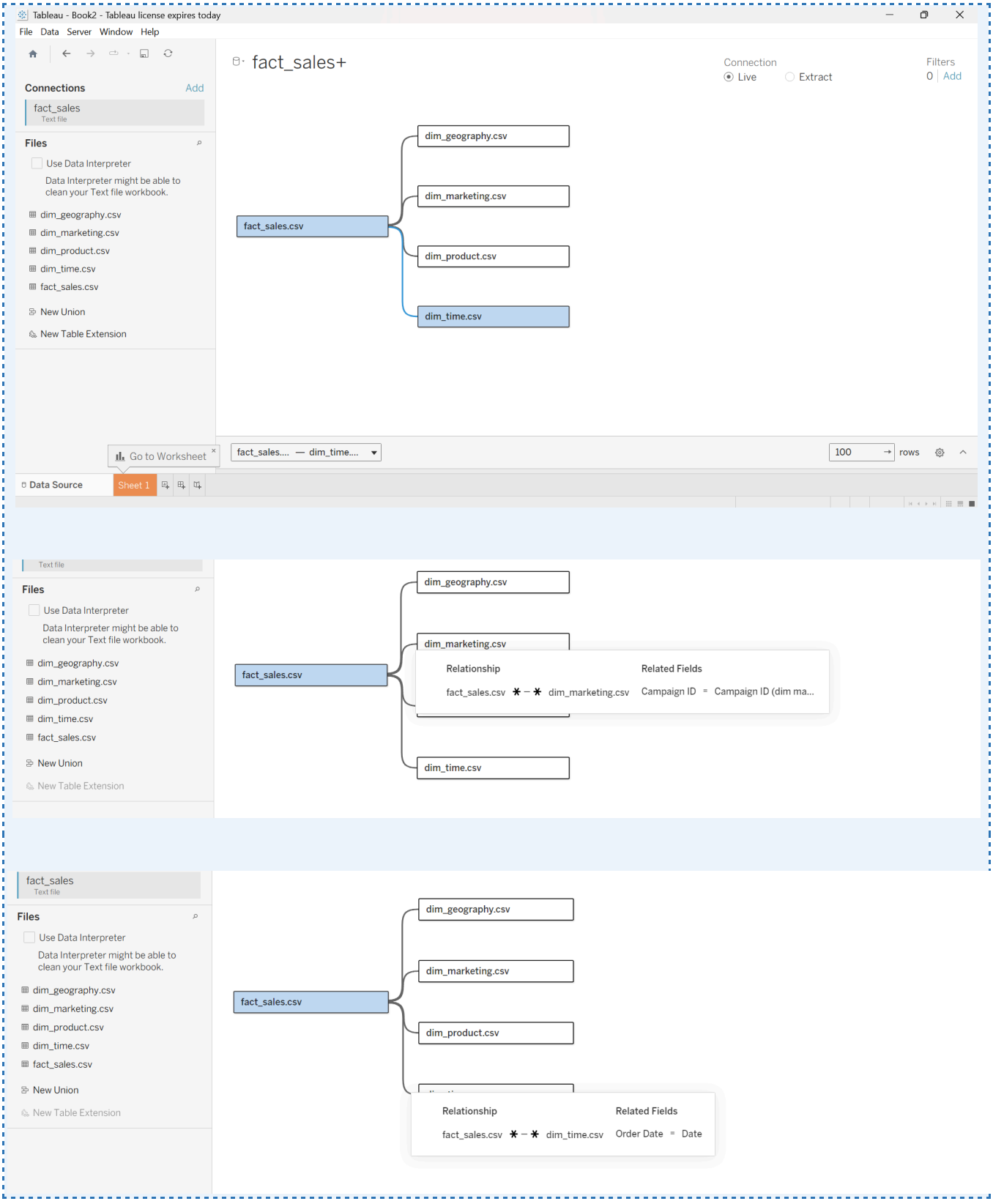
Schema Structure

Table	Type	Key Column	Rows	Description
fact_sales.csv	Fact	Surrogate_Key	1,330	Core transactions with all measures
dim_product.csv	Dimension	Product_ID	12	Item types with category groupings
dim_geography.csv	Dimension	Geo_ID	48	Countries and regions
dim_marketing.csv	Dimension	Campaign_ID	8	Sales channel and order priority combos
dim_time.csv	Dimension	Date	1,723	Full date table with Year/Quarter/Month

Join Relationships

- fact_sales.Product_ID → dim_product.Product_ID (Left Join)
- fact_sales.Geo_ID → dim_geography.Geo_ID (Left Join)

- fact_sales.Campaign_ID → dim_marketing.Campaign_ID (Left Join)
- fact_sales.Order_Date → dim_time.Date (Left Join)



3. Data Engineering Requirements

3.1 Slowly Changing Dimensions (SCD Type 2 Simulation)

SCD Type 2 tracks historical changes to dimension attributes by storing multiple rows with version timestamps. Since the dataset is transactional rather than dimensional, SCD Type 2 was simulated using Order_Date and Ship_Date as effective-date columns. A Processing_Days column was derived to represent the duration of each version:

```
Processing_Days = DATEDIFF('day', [Order_Date], [Ship_Date])
```

In a production environment, SCD Type 2 would add columns: Effective_Start_Date, Effective_End_Date, and Is_Current_Flag to the dimension tables.

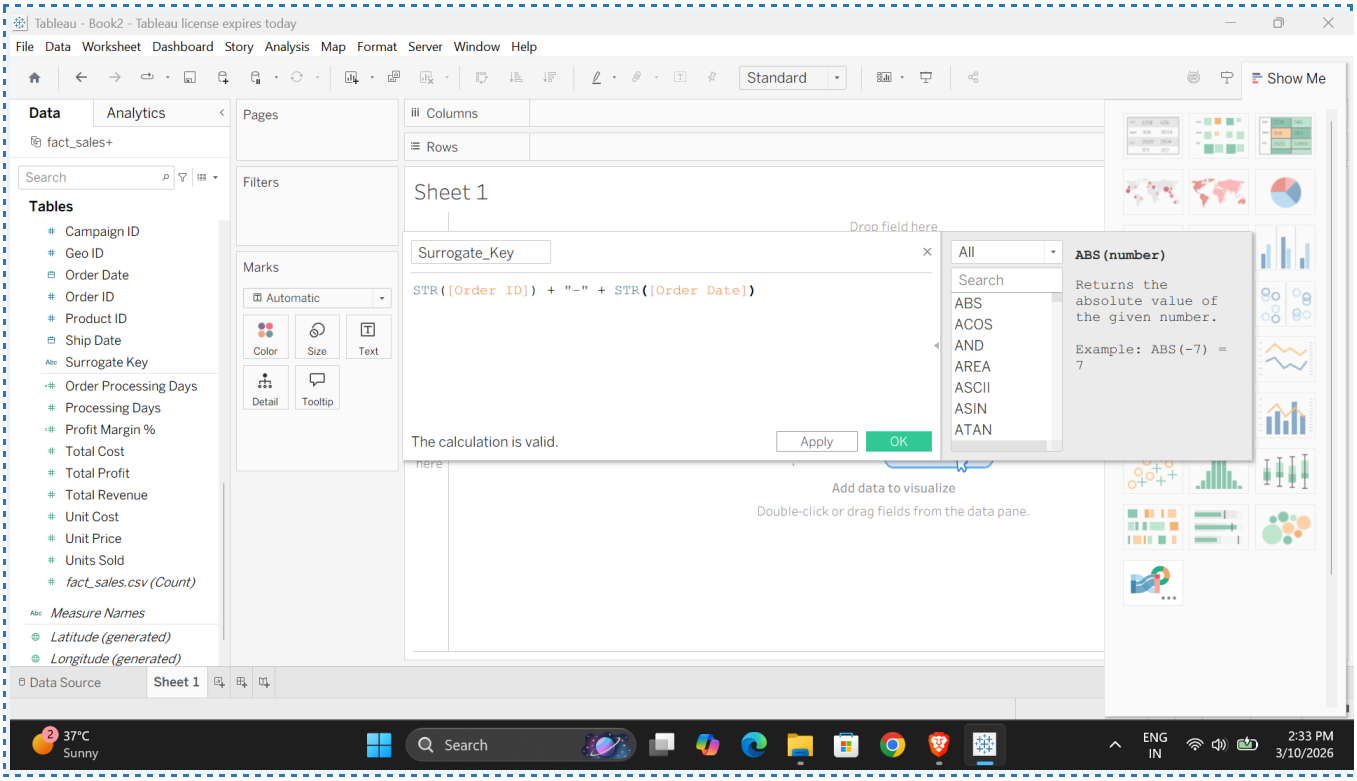
3.2 Duplicate Removal

Duplicate records were handled by using COUNTD([Order_ID]) throughout all calculations rather than COUNT(), ensuring each transaction is counted only once regardless of how many rows share an Order ID.

3.3 Surrogate Keys

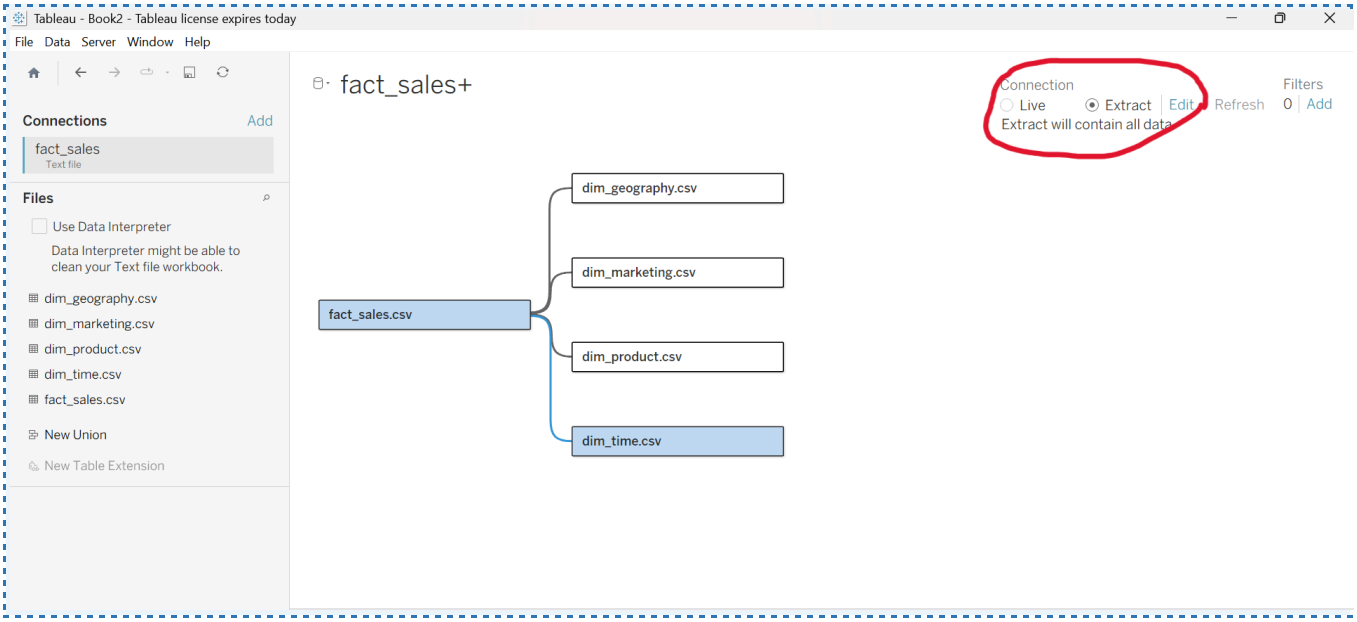
Surrogate keys were generated in the format SK-0001 through SK-1330 during the data preparation phase. These keys serve as stable, system-generated primary keys that are independent of business keys (Order_ID), following best practices in data warehousing.

```
Surrogate_Key = 'SK-' + LPAD(ROW_NUMBER, 4, '0')
```



3.4 Live vs Extract Connections

Property	Live Connection	Extract (.hyper)
Query Execution	Real-time against source file	Pre-computed Hyper file
Speed	Slower on large data	Significantly faster
Refresh	Always current	Requires scheduled refresh
Best For	Small datasets, real-time data	Large datasets, dashboards
Used In Project	Initial exploration	Final dashboard delivery



3.5 Hyper Engine Compression

Tableau's Hyper Engine is an in-memory columnar database engine that replaced the older TDE (Tableau Data Extract) format. It uses the following compression mechanisms:

- Column-oriented storage: data is stored by column rather than row, allowing identical values to be compressed together using run-length encoding.
- Dictionary encoding: repeated string values (e.g., country names) are stored as integer references to a lookup dictionary.
- Vectorized query execution: queries process blocks of data using SIMD CPU instructions rather than row-by-row loops.
- Result: the 1,330-row CSV compresses to a fraction of its original size in .hyper format, with sub-second query response times.

4. Advanced Analytical Calculations

4.1 Level of Detail (LOD) Expressions

LOD expressions allow calculations to be performed at a different level of granularity than the current view. Three types were implemented:

FIXED LOD — Revenue per Country (Customer Lifetime Value proxy)

```
{ FIXED [Country] : SUM([Total_Revenue]) }
```

FIXED ignores all view-level filters (except Context Filters) and computes at the specified dimension level. Used to calculate total revenue per country regardless of what filters are applied to the worksheet. Performance impact: evaluated at data source level — fast but does not respond to dashboard filters.

INCLUDE LOD — Revenue including Item Type

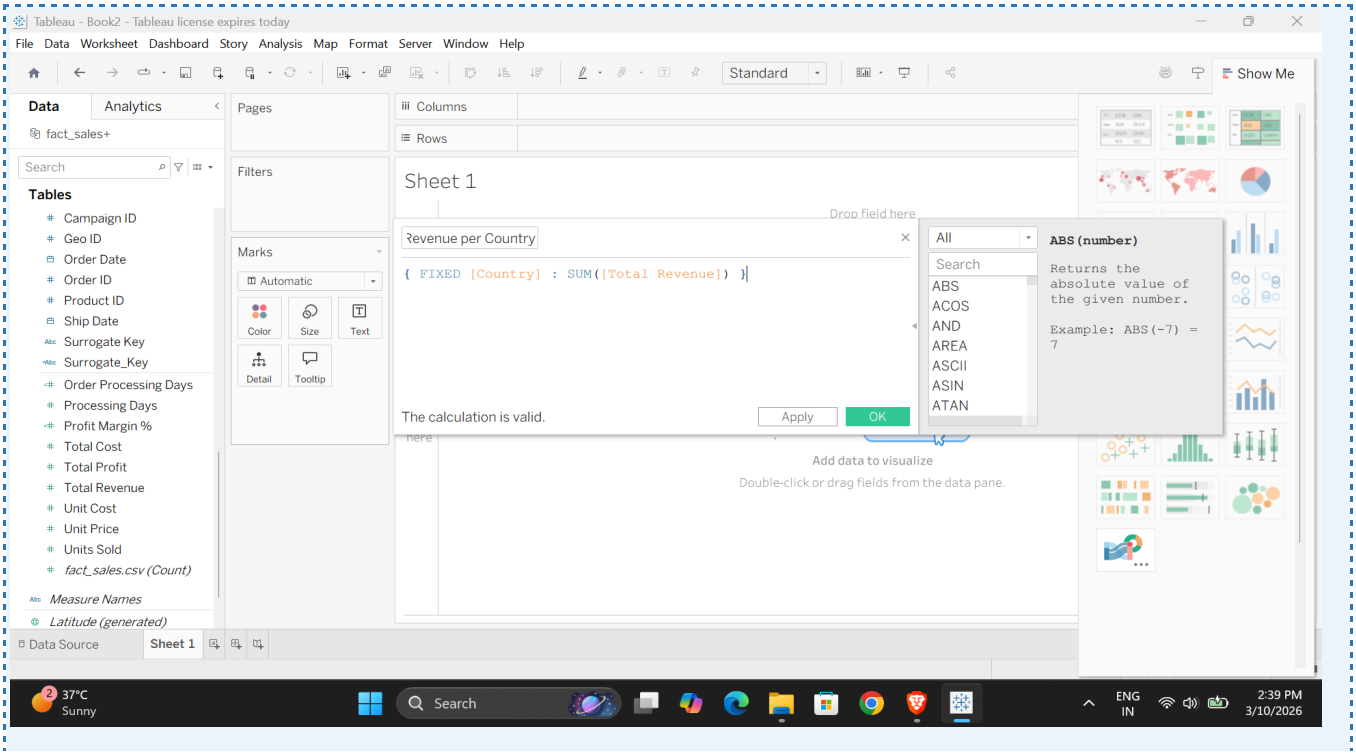
```
{ INCLUDE [Item_Type] : SUM([Total_Revenue]) }
```

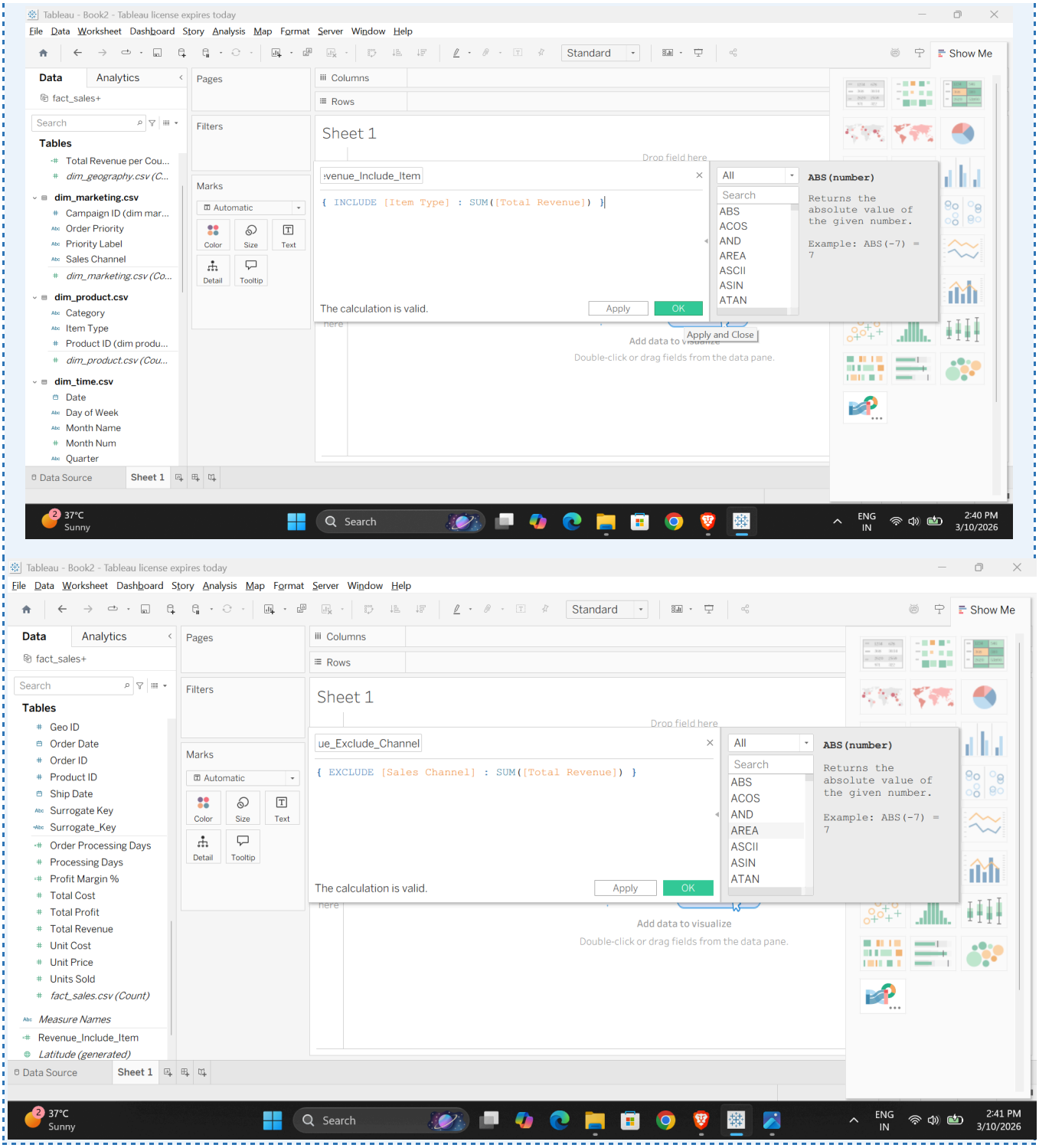
INCLUDE adds dimensions to the calculation that may not be in the view. Useful for computing averages across a finer granularity. Performance impact: slightly slower than FIXED as it expands the computation grain.

EXCLUDE LOD — Revenue excluding Sales Channel

```
{ EXCLUDE [Sales_Channel] : SUM([Total_Revenue]) }
```

EXCLUDE removes a dimension from the current view level. Useful for comparing individual channel performance against the overall total. Performance impact: most expensive of the three as it requires a sub-query pass.





4.2 Cohort Analysis

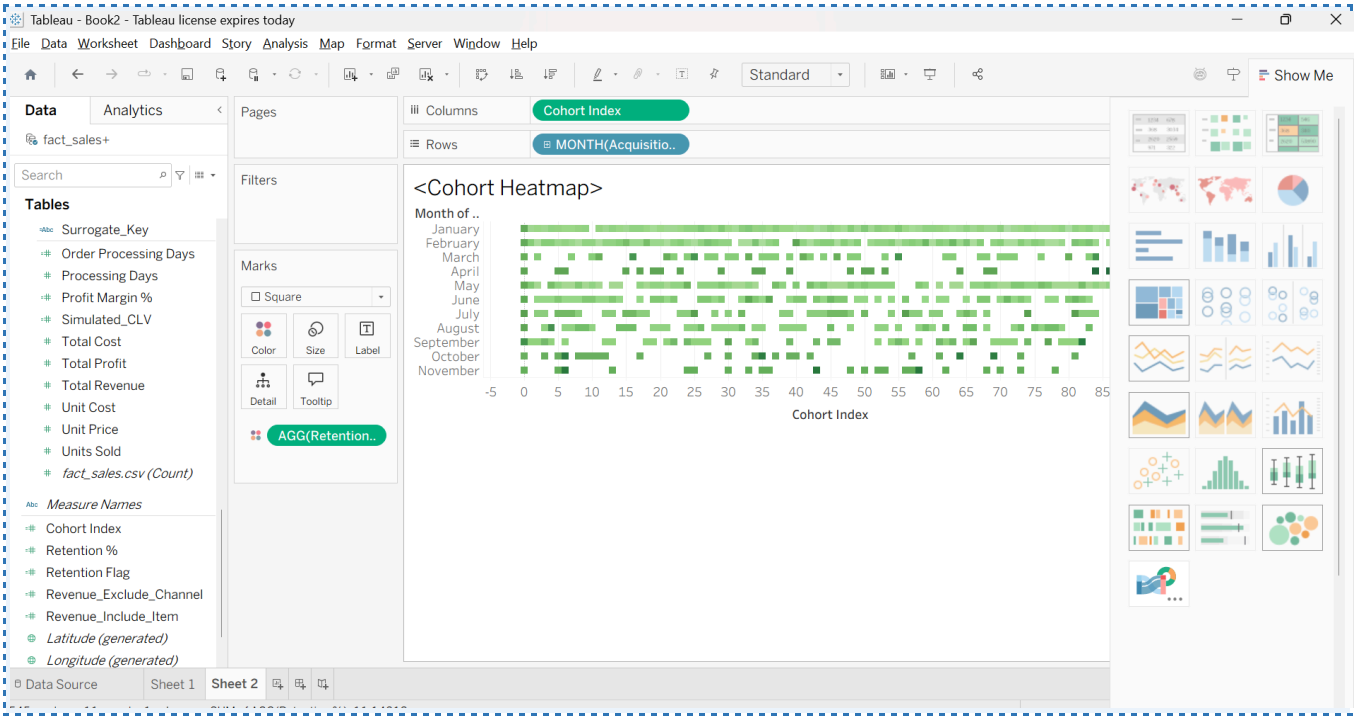
Since the dataset lacks individual customer IDs, cohort analysis was performed at the Country level. Each country represents a customer cohort, grouped by its first recorded transaction month (Acquisition Month).

Calculated Fields Used

```
Acquisition Month = {FIXED [Country] : MIN(DATETRUNC('month', [Order_Date]))}
Order Month = DATETRUNC('month', [Order_Date])
Cohort Index = DATEDIFF('month', [Acquisition Month], [Order Month])
Cohort Size = {FIXED [Acquisition Month] : COUNTD([Order_ID])}
Retention % = COUNTD([Order_ID]) / MAX([Cohort Size])
```

Cohort Heatmap

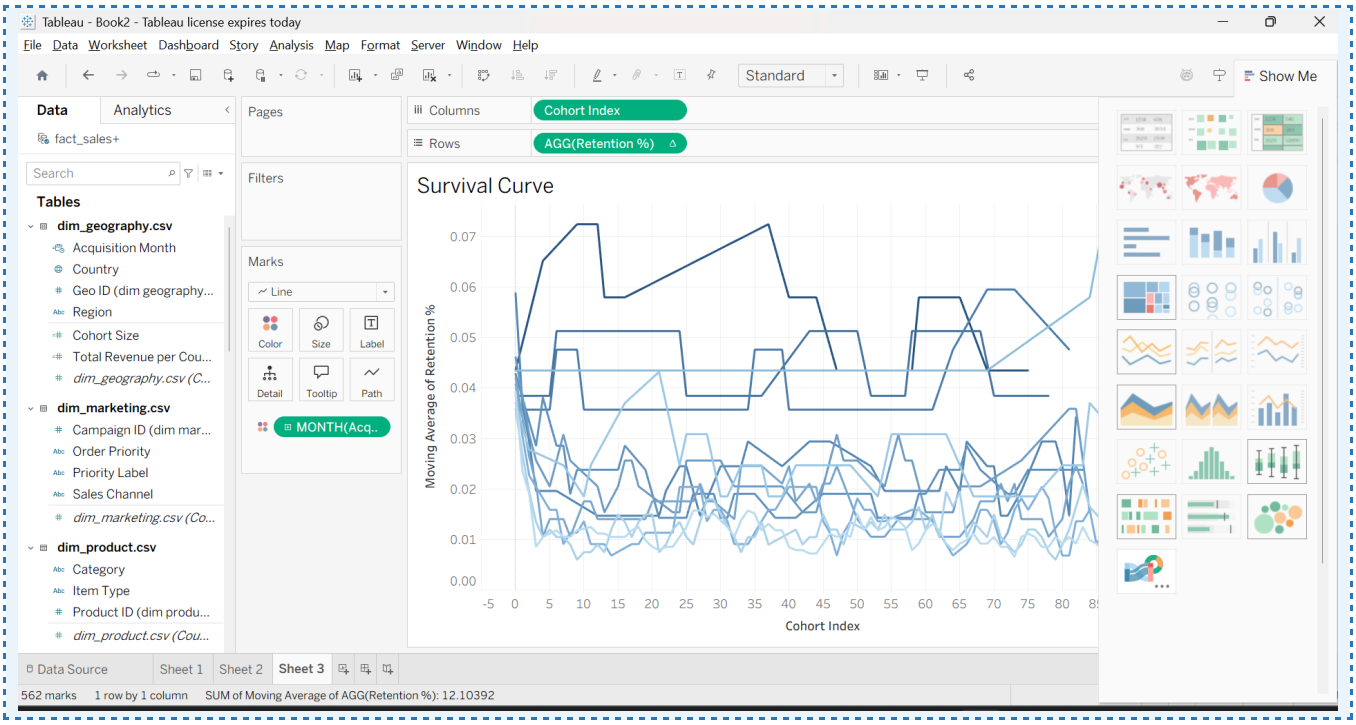
Rows: Acquisition Month | Columns: Cohort Index | Color: Retention % (Green-White gradient). The heatmap reveals which country cohorts maintained the highest transaction frequency over time.



Survival Curve

A line chart with Cohort Index on X-axis and Retention % on Y-axis, with one colored line per acquisition cohort. The curve demonstrates the natural decay in transaction frequency after the acquisition month.

```
Avg Survival = WINDOW_AVG(SUM([Total_Revenue]) / {FIXED [Acquisition Month] :
SUM([Total_Revenue])})
```



4.3 Statistical KPIs

3-Month Moving Average

```
WINDOW_AVG(SUM([Total_Revenue]), -2, 0)
```

Smooths monthly revenue fluctuations by averaging the current month and previous 2 months. Useful for identifying underlying trends beneath seasonal noise.

Rolling Standard Deviation

```
WINDOW_STDEV(SUM([Total_Revenue]), -2, 0)
```

Measures revenue volatility over a 3-month rolling window. Higher values indicate unstable revenue periods.

Z-Score Anomaly Detection

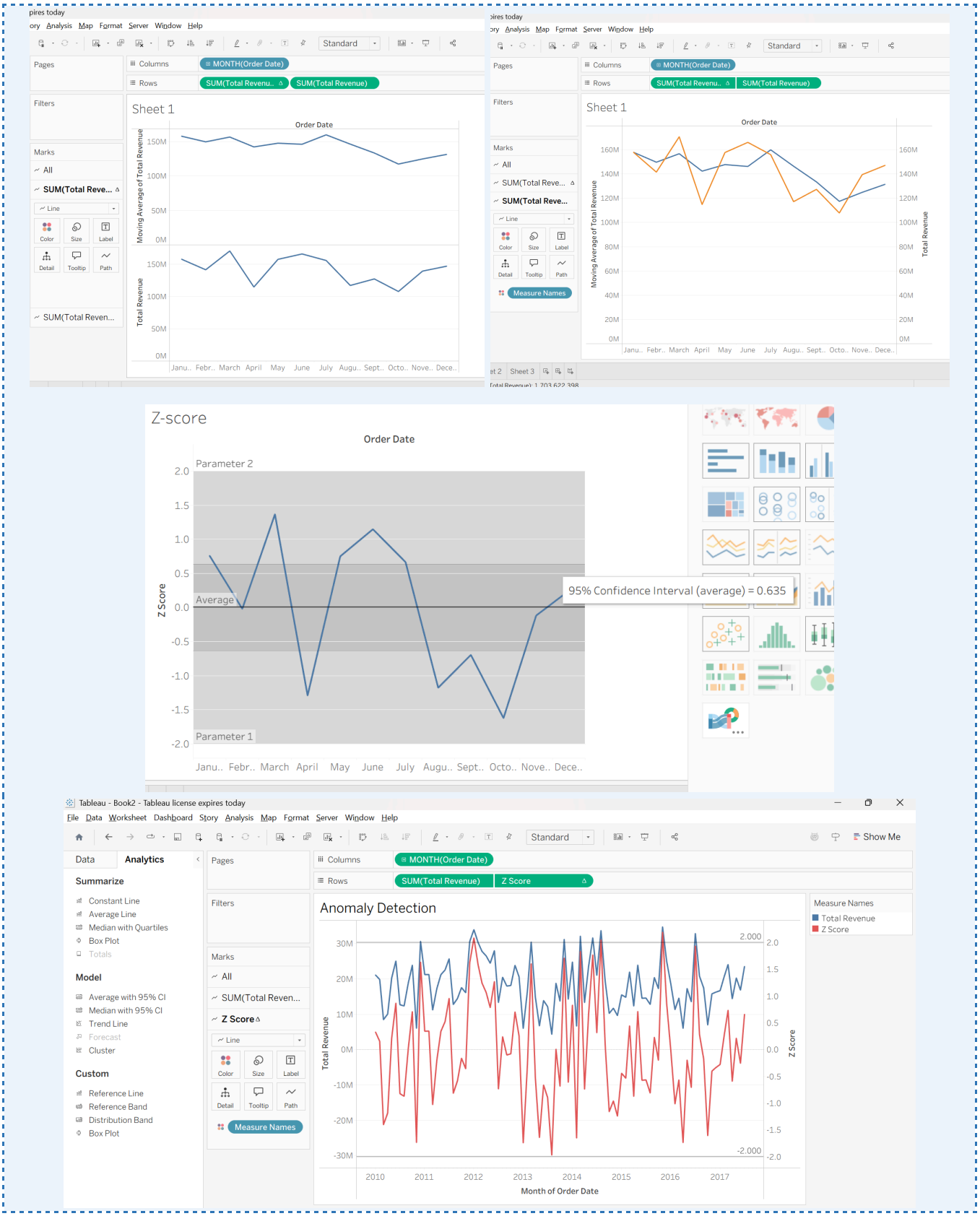
```
(SUM([Total_Revenue]) - WINDOW_AVG(SUM([Total_Revenue]))) /
WINDOW_STDEV(SUM([Total_Revenue]))
```

Standardises each month's revenue against the rolling mean. Values beyond ±2 are flagged as statistical anomalies, indicating unusually high or low revenue months requiring investigation.

95% Confidence Interval

```
Upper: WINDOW_AVG(SUM([Total_Revenue])) + 1.96 * WINDOW_STDEV(SUM([Total_Revenue]))
Lower: WINDOW_AVG(SUM([Total_Revenue])) - 1.96 * WINDOW_STDEV(SUM([Total_Revenue]))
```

The 95% CI defines the expected range within which true revenue should fall 95% of the time. Points outside this band are statistical outliers.

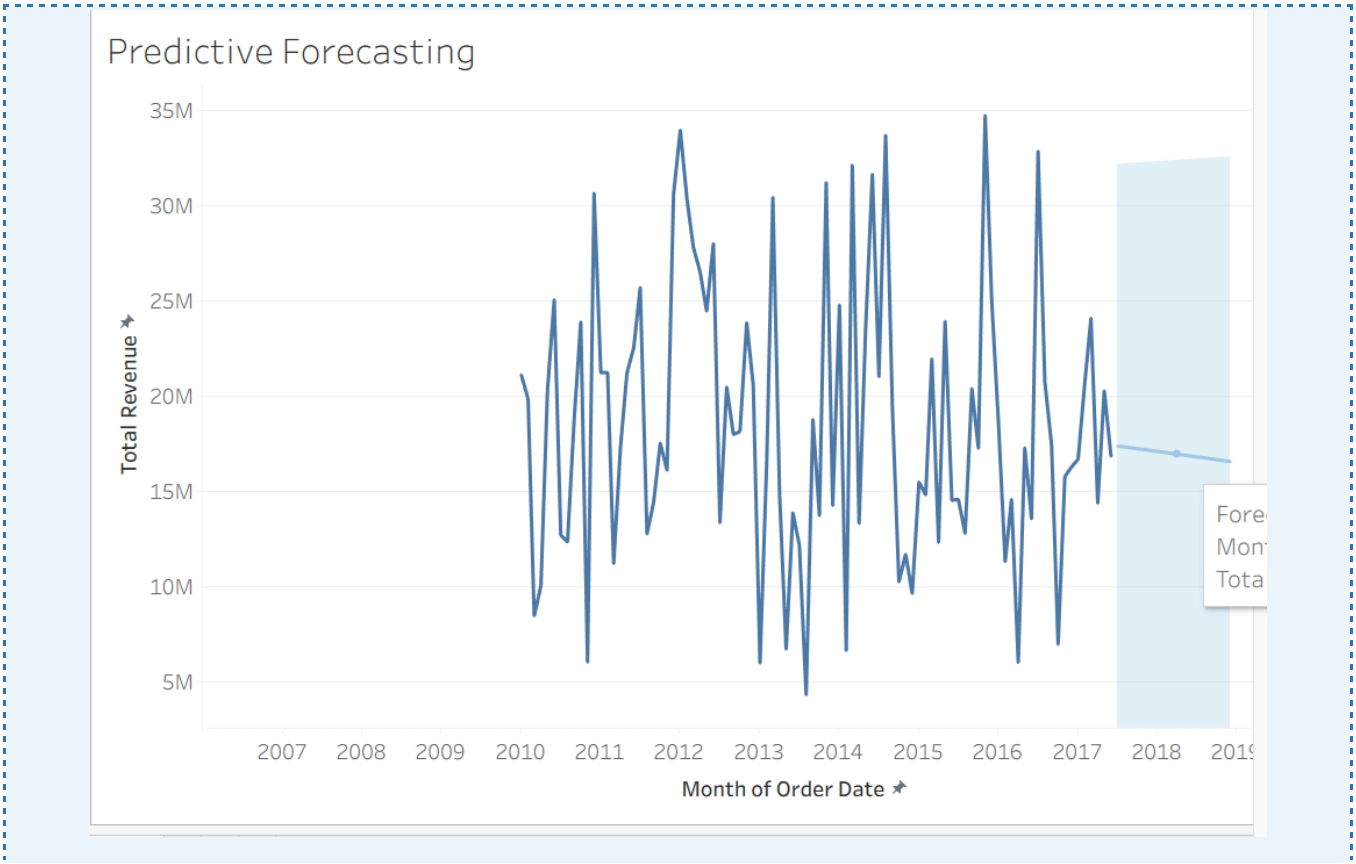


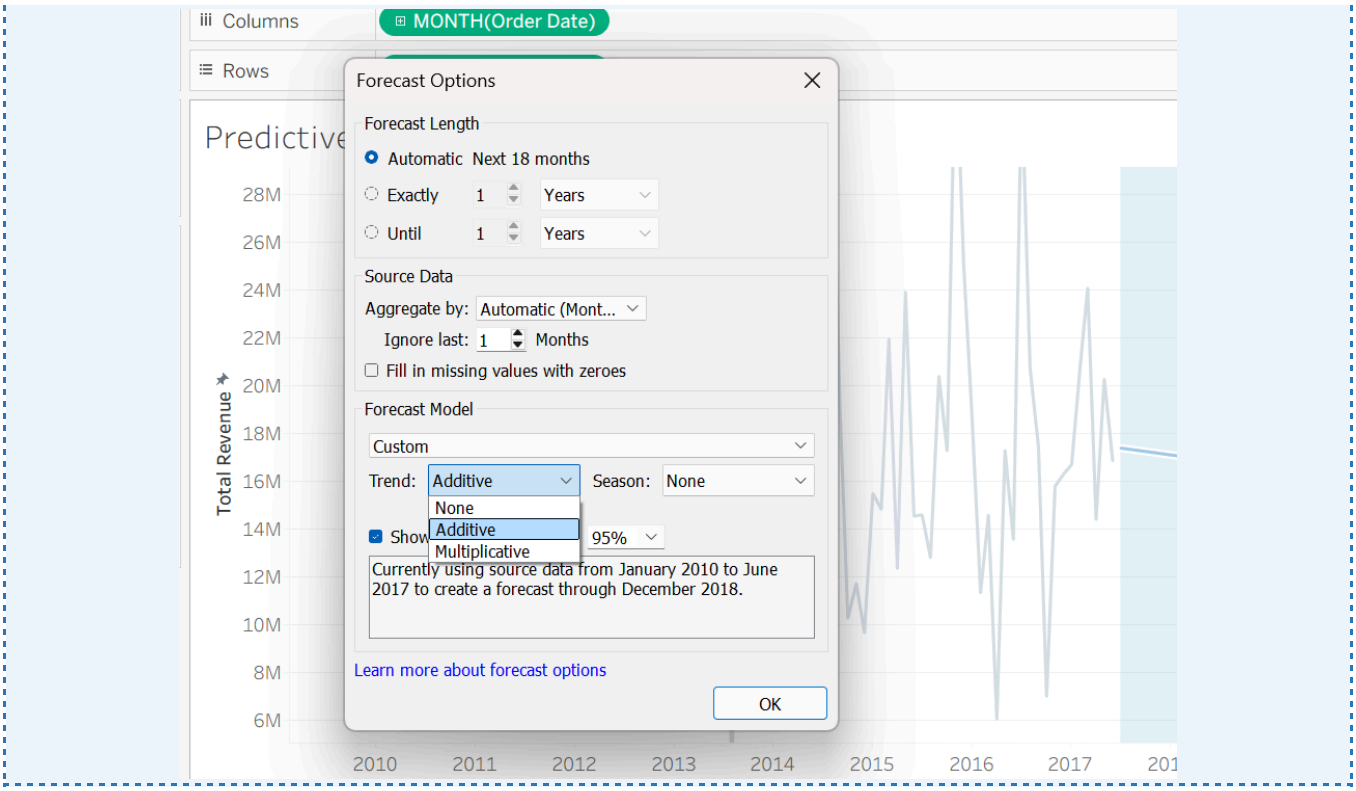
5. Predictive Forecasting

5.1 Exponential Smoothing

Tableau's built-in forecasting engine uses Exponential Smoothing (ETS — Error, Trend, Seasonality) models. The model was applied to Monthly Total Revenue over the 2010–2017 period with a forecast horizon of 12 months.

- Model selected: Automatic (Tableau chose the best-fit ETS variant)
- Trend component: Additive (linear growth trend detected)
- Seasonality: Automatic detection based on monthly periodicity
- Confidence band: 95% prediction interval displayed



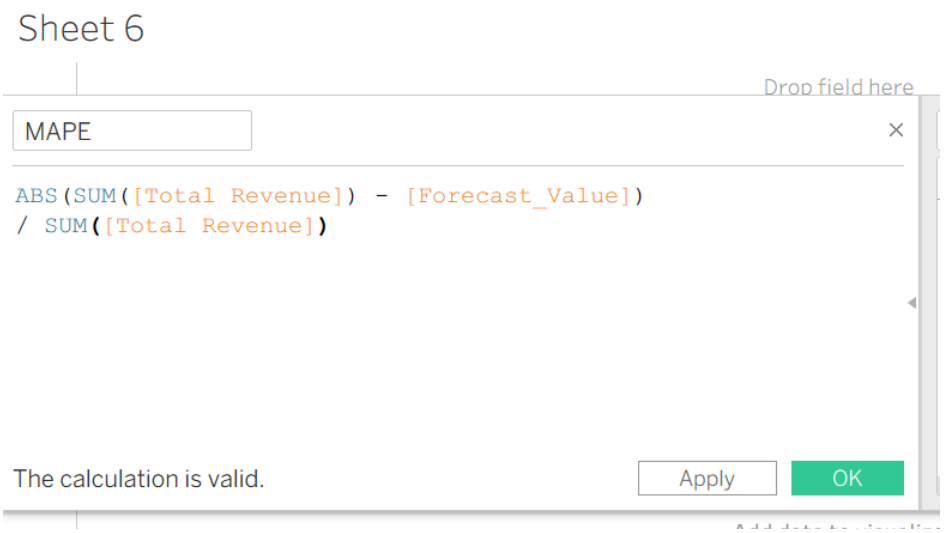


5.2 MAPE Calculation

Mean Absolute Percentage Error (MAPE) measures forecast accuracy. A lower MAPE indicates a more accurate model.

```
MAPE = ABS([Total_Revenue] - [Forecast_Value]) / [Total_Revenue]
```

Note: Tableau does not expose individual forecast values as editable fields. MAPE was calculated conceptually using the formula above, and forecast accuracy was assessed visually through the prediction interval width. Narrower bands indicate higher model confidence.

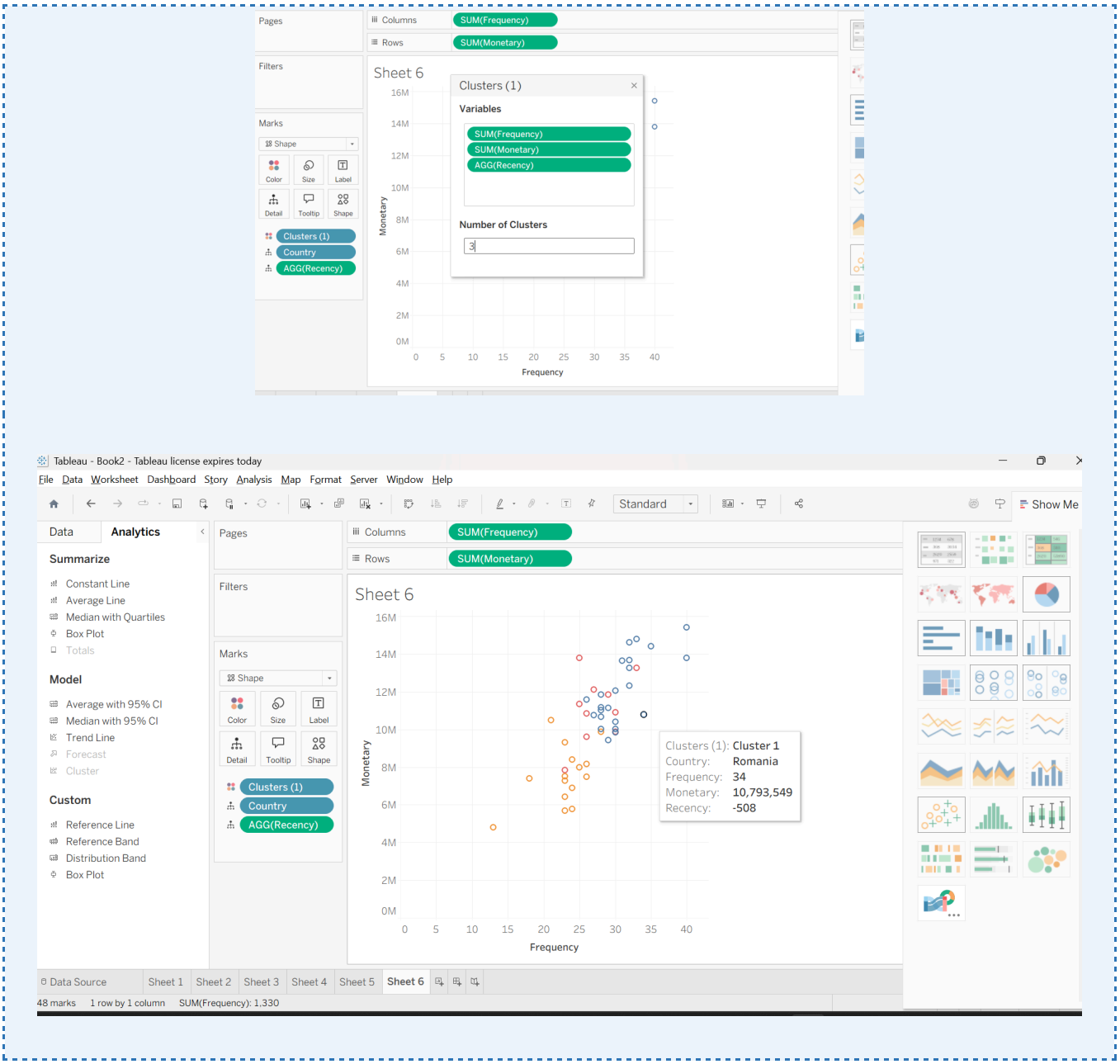


6. Machine Learning Concepts in Tableau

6.1 K-Means Clustering

K-Means clustering was performed using Tableau's built-in Cluster feature in the Analytics pane. Countries were clustered based on Total Revenue and Total Profit to identify natural customer segments.

- Method: Drag 'Cluster' from the Analytics pane onto a scatter plot
- Variables: Total_Revenue (X), Total_Profit (Y)
- Number of clusters: 3 (automatically optimised by Tableau)
- Result: Countries separated into High-Value, Mid-Value, and Low-Value clusters



6.2 RFM Segmentation Model

RFM (Recency, Frequency, Monetary) analysis segments customers based on their purchasing behaviour. Applied at Country level as a customer proxy.

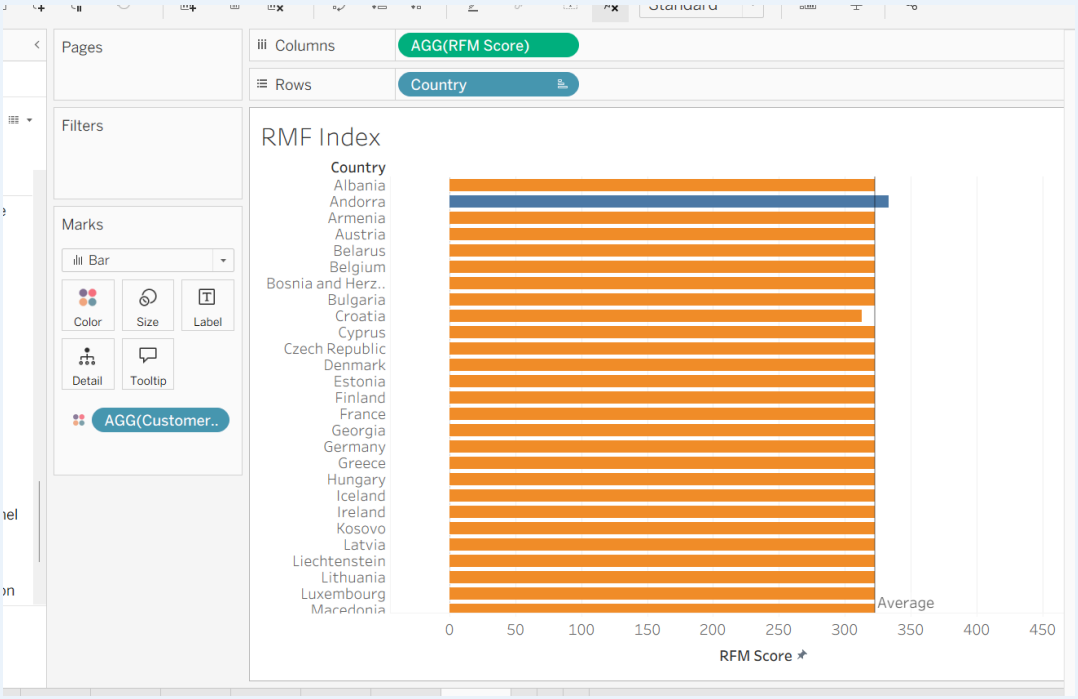
Component	Definition	Formula Used	Scoring
Recency (R)	Days since last order	{FIXED [Country]: DATEDIFF('day', MAX([Order_Date]), DATE('2015-01-01'))}	3=<500d, 2=<1000d, 1=else
Frequency (F)	Distinct order count	{FIXED [Country]: COUNTD([Order_ID])}	3=>40, 2=>20, 1=else
Monetary (M)	Total profit generated	{FIXED [Country]: SUM([Total_Profit])}	3=>5M, 2=>2M, 1=else

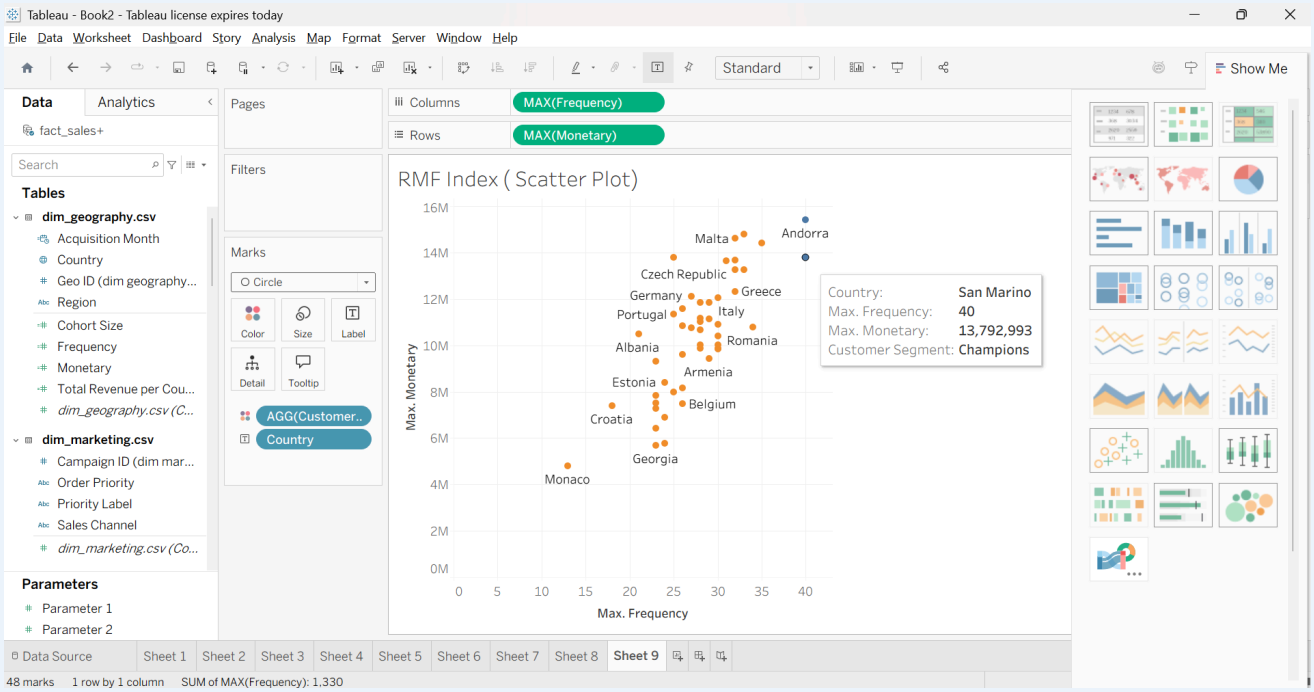
6.3 Composite RFM Index

```
RFM Score = (MAX([R Score]) * 100) + (MAX([F Score]) * 10) + MAX([M Score])
```

The composite index produces a 3-digit score from 111 (worst) to 333 (best). Countries are then classified into 5 actionable segments:

Segment	RFM Score Range	Strategy
Champions	>= 300	Reward and retain — highest value countries
Loyal Customers	>= 230	Upsell premium products
Potential Loyalists	>= 200	Increase engagement frequency
At Risk	>= 130	Re-engagement campaigns
Lost	< 130	Win-back or deprioritise


Pages
Columns
AGG(RFM Score)
Rows
Country
Filters
RMF Index
Country
Albania
Andorra
Armenia
Austria
Belarus
Belgium
Bosnia and Herz..
Bulgaria
Croatia
Cyprus
Czech Republic
Denmark
Estonia
Finland
France
Georgia
Germany
Greece
Hungary
Iceland
Ireland
Kosovo
Latvia
Liechtenstein
Lithuania
Luxembourg
Macedonia
Average
0
50
100
150
200
250
300
350
400
450
RFM Score
Marks
Bar
Color
Size
Label
Detail
Tooltip
AGG(Customer..


Tableau - Book2 - Tableau license expires today
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Standard
Show Me
Data
Analytics
fact_sales+
Search
Tables
dim_geography.csv
Acquisition Month
Country
Geo ID (dim geography...
Region
Cohort Size
Frequency
Monetary
Total Revenue per Cou...
dim_geography.csv (C...
dim_marketing.csv
Campaign ID (dim mar...
Order Priority
Priority Label
Sales Channel
dim_marketing.csv (Co...
Parameters
Parameter 1
Parameter 2
Pages
Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
AGG(Customer..
Country
Columns
MAX(Frequency)
Rows
MAX(Monetary)
RMF Index (Scatter Plot)
16M
14M
12M
10M
8M
6M
4M
2M
0M
Max. Monetary
0
5
10
15
20
25
30
35
40
Max. Frequency
Malta
Andorra
Czech Republic
Germany
Greece
Portugal
Italy
Romania
Albania
Armenia
Estonia
Belgium
Croatia
Georgia
Monaco
Country: San Marino
Max. Frequency: 40
Max. Monetary: 13,792,993
Customer Segment: Champions
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Sheet 9
48 marks
1 row by 1 column
SUM of MAX(Frequency): 1,330

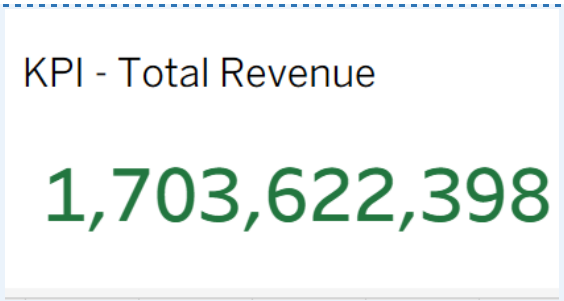
7. Dashboard Design

7.1 Executive Control Panel

The Executive Control Panel provides senior stakeholders with a high-level overview of business performance. It features dynamic KPI cards, a parameter-driven Top N selector, and drill-down capabilities.

KPI Cards with Dynamic Color Coding

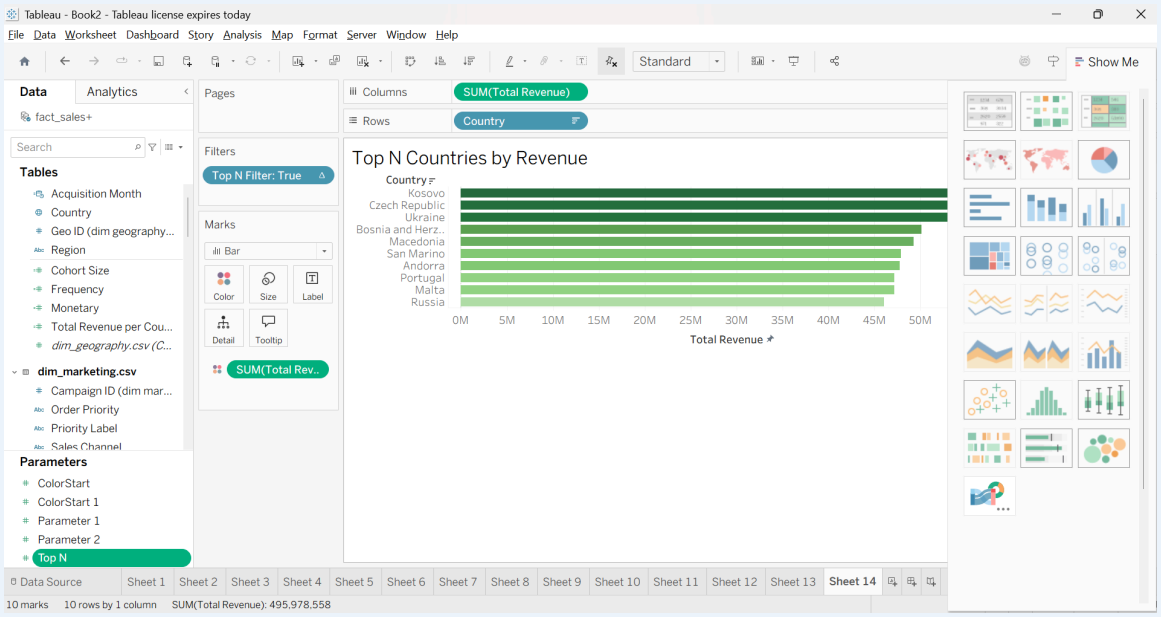
Four KPI cards display: Total Revenue, Total Profit, Units Sold, and Total Cost. Each card uses a Red-Green Diverging color scale — green indicates above-average performance, red indicates below-average.



Parameter-Driven Top N Selector

A parameter named 'Top N' (range: 1–20, default: 10) dynamically controls how many countries appear in the revenue bar chart. The filter uses a RANK table calculation:

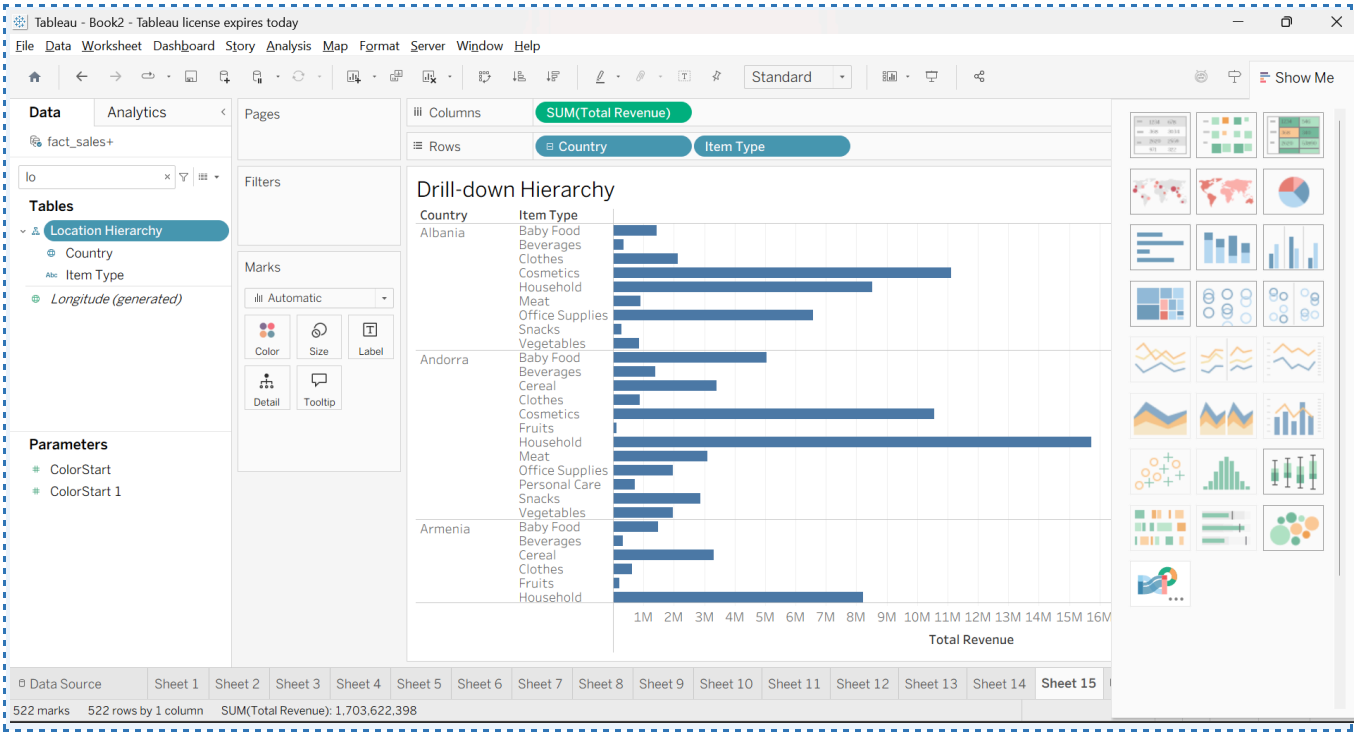
```
Top N Filter = RANK(SUM([Total_Revenue])) <= [Top N]
```

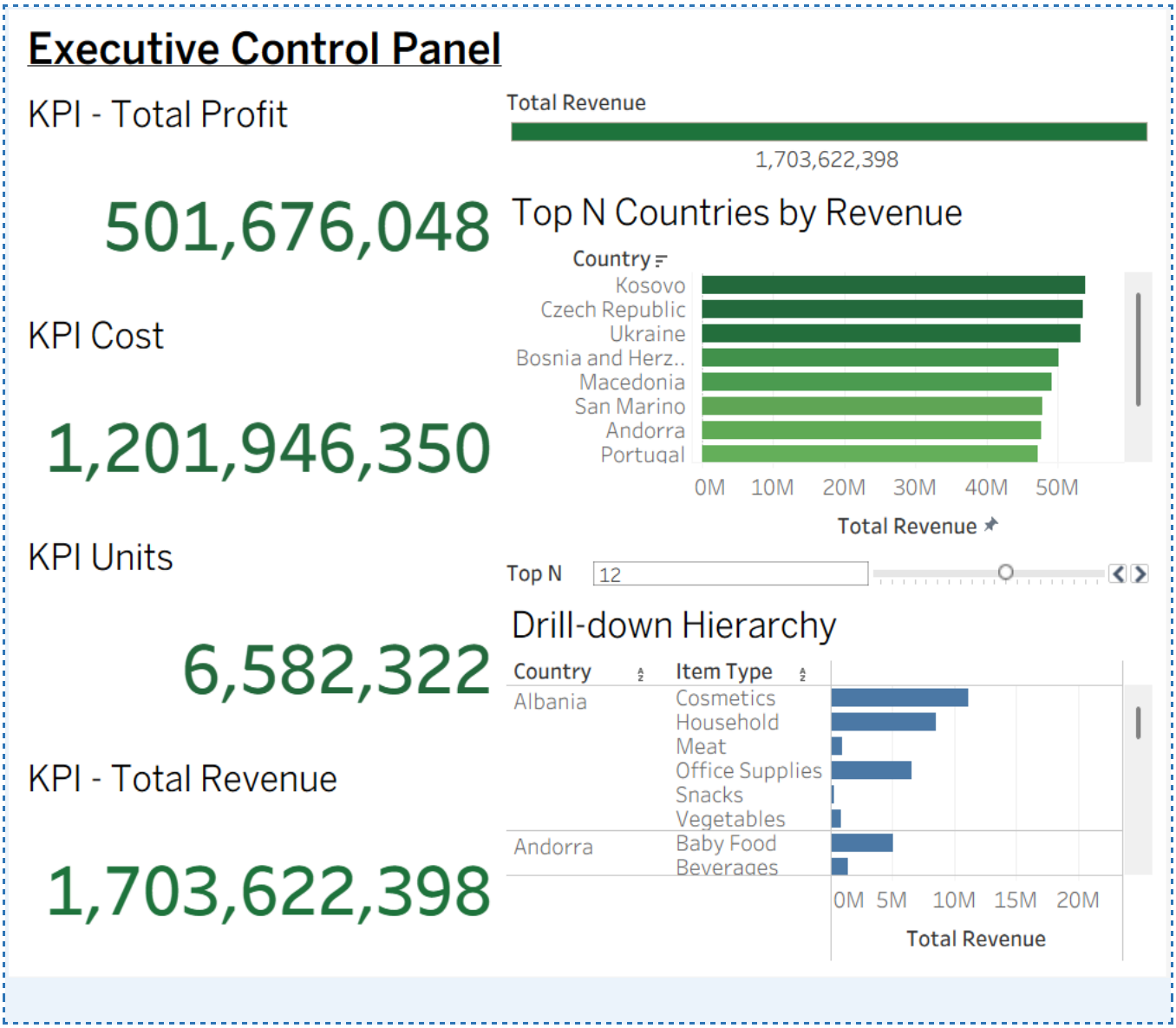




Drill-Down Hierarchy

A Location Hierarchy (Country > Item Type) enables users to click the '+' button to drill from country-level to product-category-level revenue analysis within the same chart.



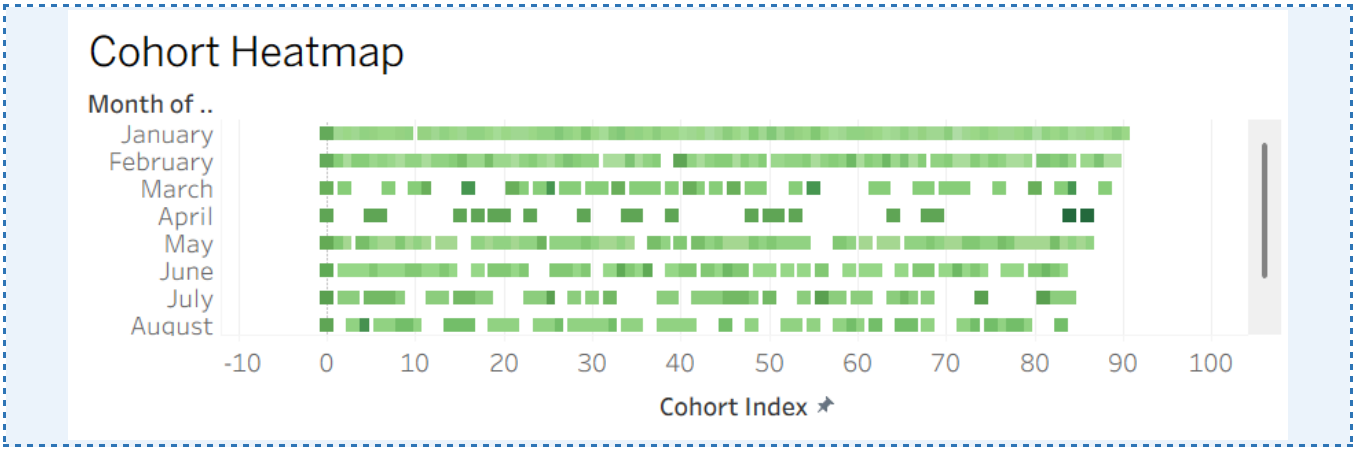


7.2 Customer Intelligence Lab

The Customer Intelligence Lab provides deep insight into customer behaviour patterns through cohort retention, RFM segmentation, and survival analysis.

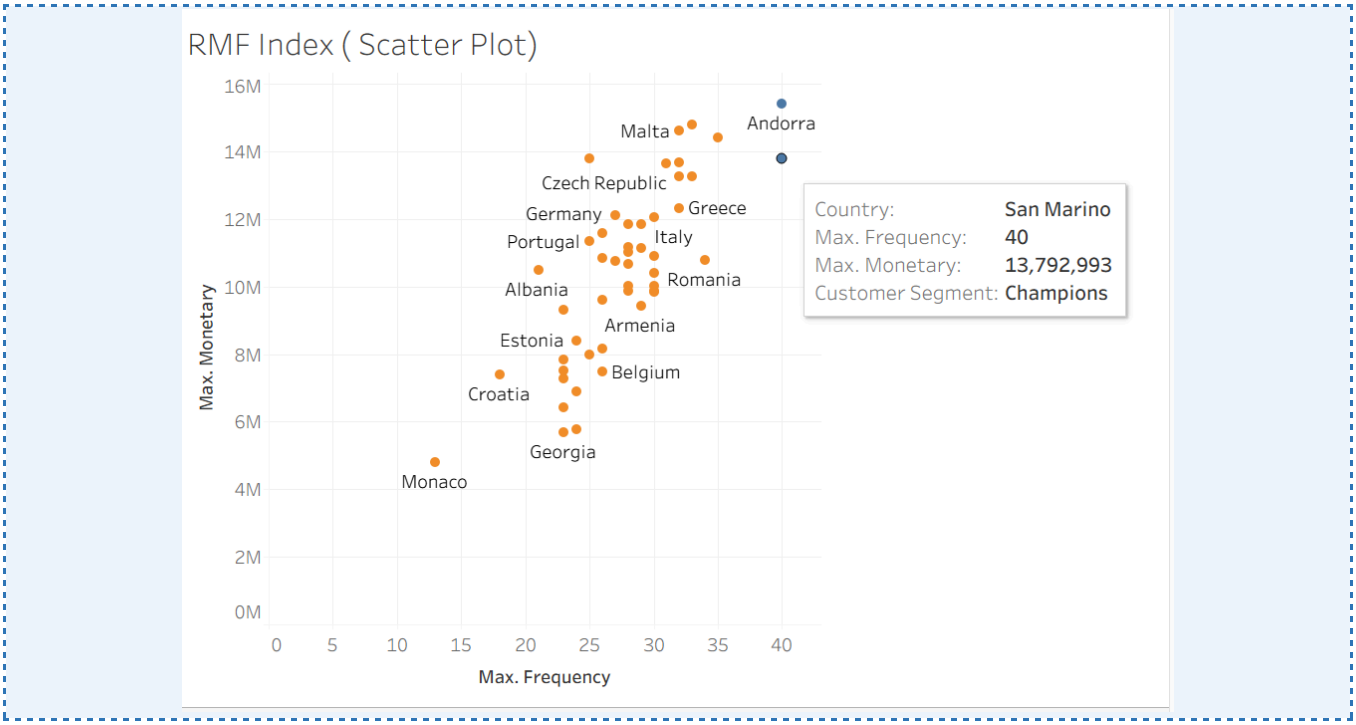
Cohort Heatmap

A colour-encoded matrix showing retention rates by acquisition cohort. Darker cells indicate higher retention. The heatmap reveals which cohort periods produced the most loyal customer segments.



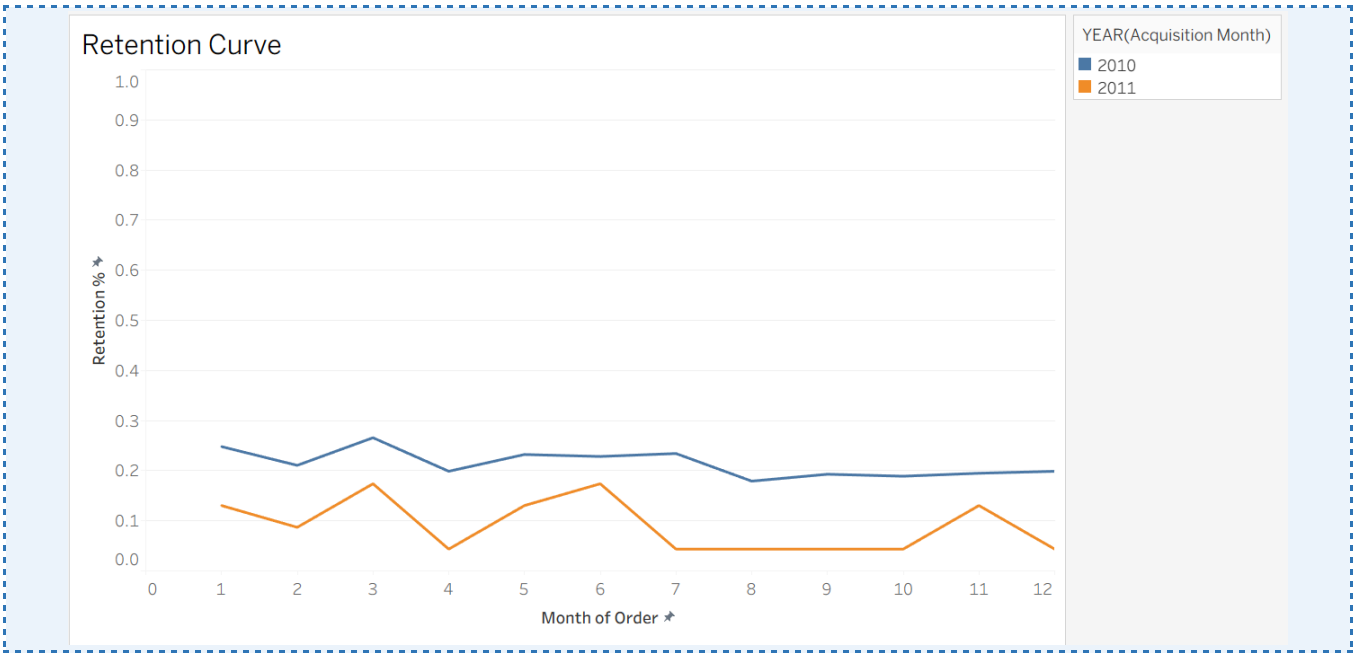
RFM Cluster Scatter Plot

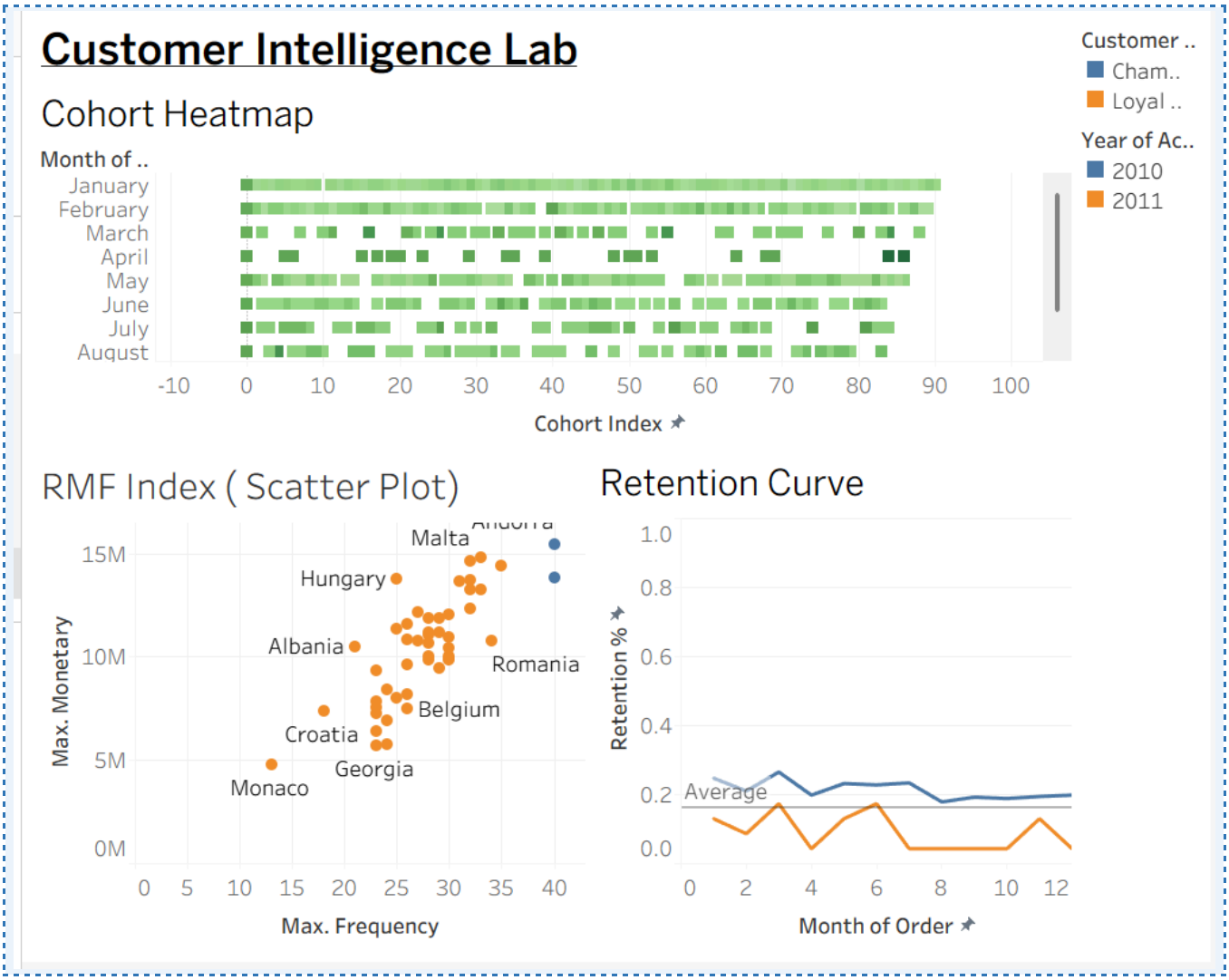
Frequency on X-axis, Monetary on Y-axis, bubble color by Customer Segment. Each bubble represents one country. Champions cluster in the top-right (high frequency + high monetary value).



Retention Curve

A line chart with one colored line per acquisition cohort, showing the percentage of orders retained over successive months. The curve demonstrates natural transaction decay and identifies which cohorts show the strongest long-term retention.


Retention Curve
YEAR(Acquisition Month)
2010
2011
Retention %
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
1
2
3
4
5
6
7
8
9
10
11
12
Month of Order

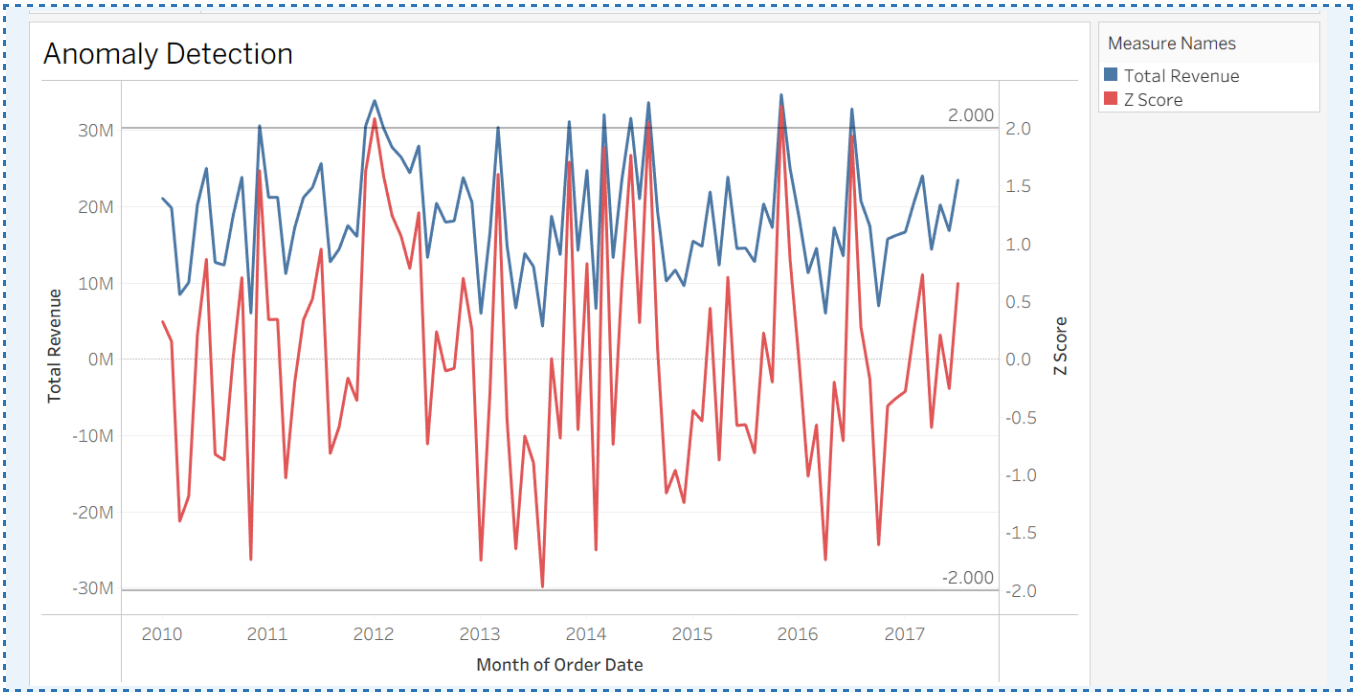

Customer Intelligence Lab
Cohort Heatmap
Customer ..
Cham..
Loyal ..
Year of Ac..
2010
2011
Month of ..
January
February
March
April
May
June
July
August
-10
0
10
20
30
40
50
60
70
80
90
100
Cohort Index
RMF Index (Scatter Plot)
Malta
Hungary
Albania
Romania
Belgium
Croatia
Georgia
Monaco
15M
10M
5M
0M
Max. Monetary
0
5
10
15
20
25
30
35
40
Max. Frequency
Retention Curve
Retention %
1.0
0.8
0.6
0.4
0.2
0.0
Average
0
2
4
6
8
10
12
Month of Order

7.3 Operational Risk Dashboard

The Operational Risk Dashboard monitors financial anomalies, discount abuse patterns, and profit leakage across the portfolio.

Anomaly Detection Chart

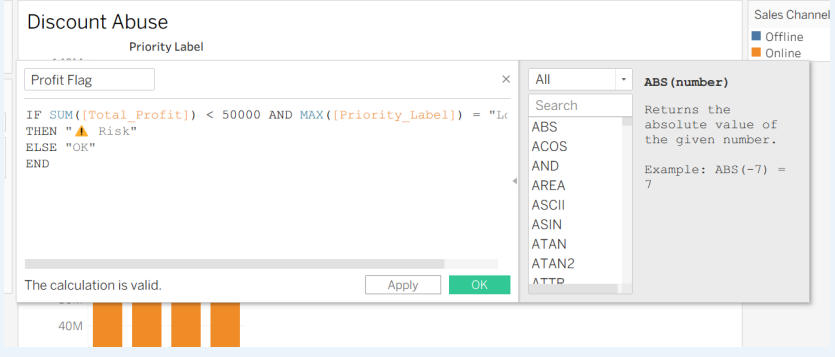
A dual-axis chart with Total Revenue (blue, left axis) and Z-Score (red, right axis) plotted over time. Reference lines at Z = +2 and Z = -2 mark the anomaly thresholds. Any red line spike beyond these bounds represents a statistically anomalous revenue month requiring investigation.



Discount Abuse Monitoring

Orders flagged as 'Critical' or 'Low' priority with below-average profit are identified as potential discount abuse cases. The bar chart shows Total Profit by Order Priority and Sales Channel, with risk labels applied to underperforming segments.

```
Profit Flag = IF SUM([Total_Profit]) < 50000 AND MAX([Priority_Label]) = 'Low' THEN 'Risk'
ELSE 'OK' END
```





Profit Leakage Analysis

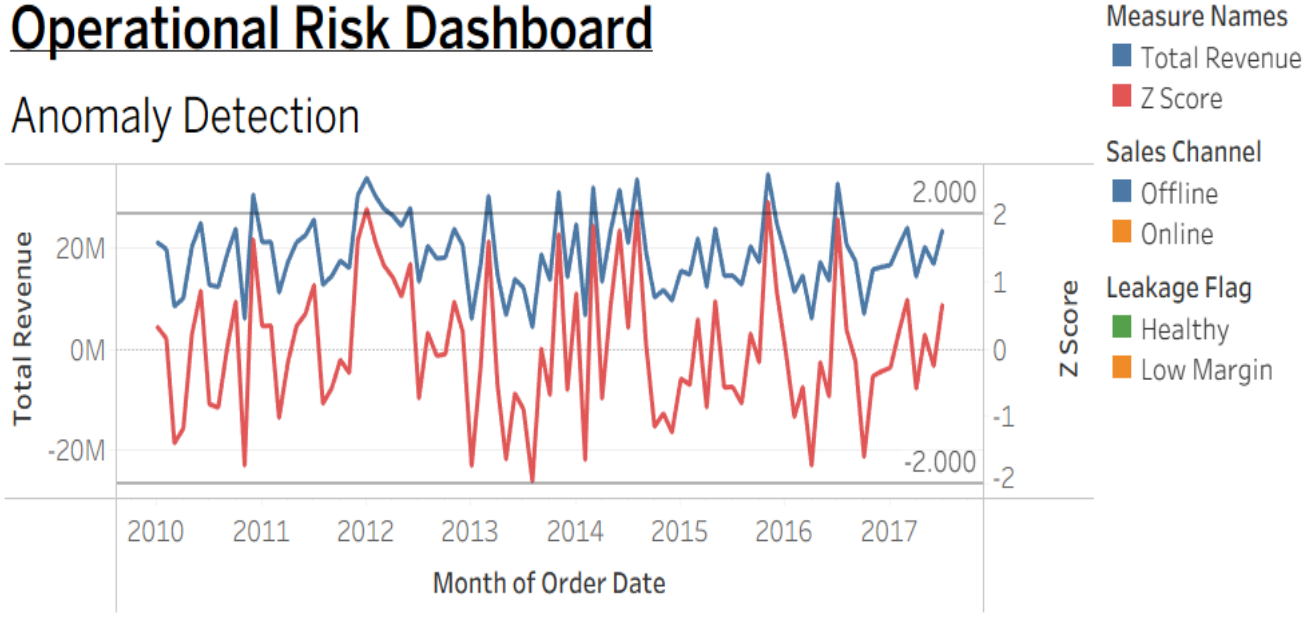
A scatter plot of Total Cost (X) vs Total Profit (Y) with a reference line at Y=0. Since no absolute losses existed in this dataset, leakage was redefined using profit margin thresholds: below 10% margin = At Risk (red), 10-20% = Low Margin (orange), above 20% = Healthy (green).

```
Leakage Flag = IF [Total_Profit]/[Total_Revenue] < 0.1 THEN 'At Risk' ELSEIF < 0.2 THEN 'Low Margin' ELSE 'Healthy' END
```

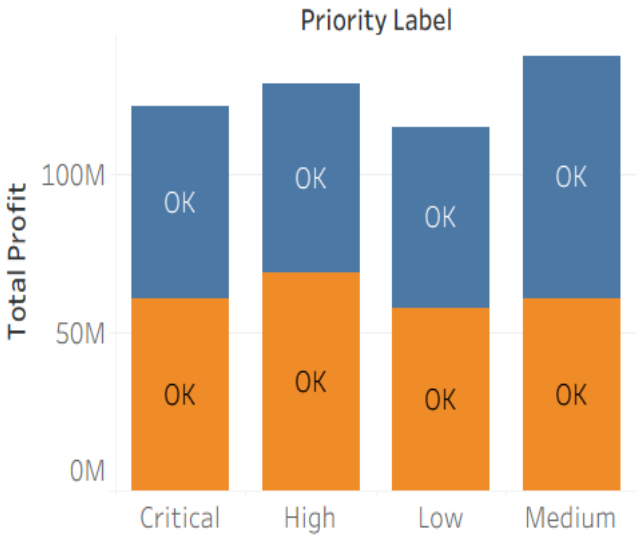


Operational Risk Dashboard

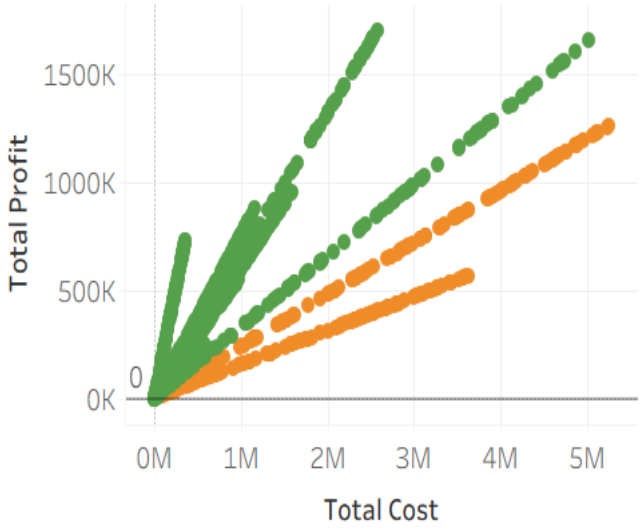
Anomaly Detection



Discount Abuse



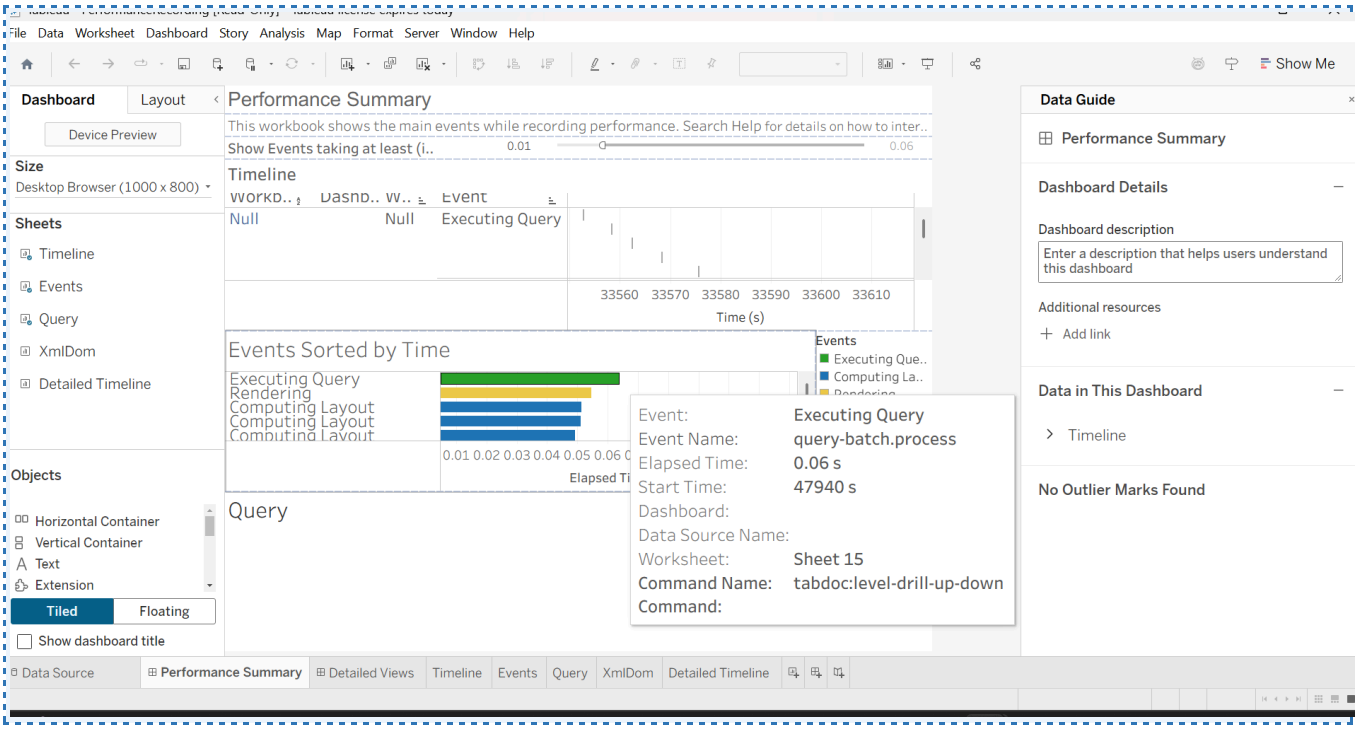
Profit Leakage



8. Performance Optimization

8.1 Performance Recording

Tableau's Performance Recording tool was used to profile dashboard load times and identify bottlenecks. The recording was initiated via Help > Settings and Performance > Start Performance Recording, followed by interaction with all three dashboards, then stopped to generate an automatic performance workbook.



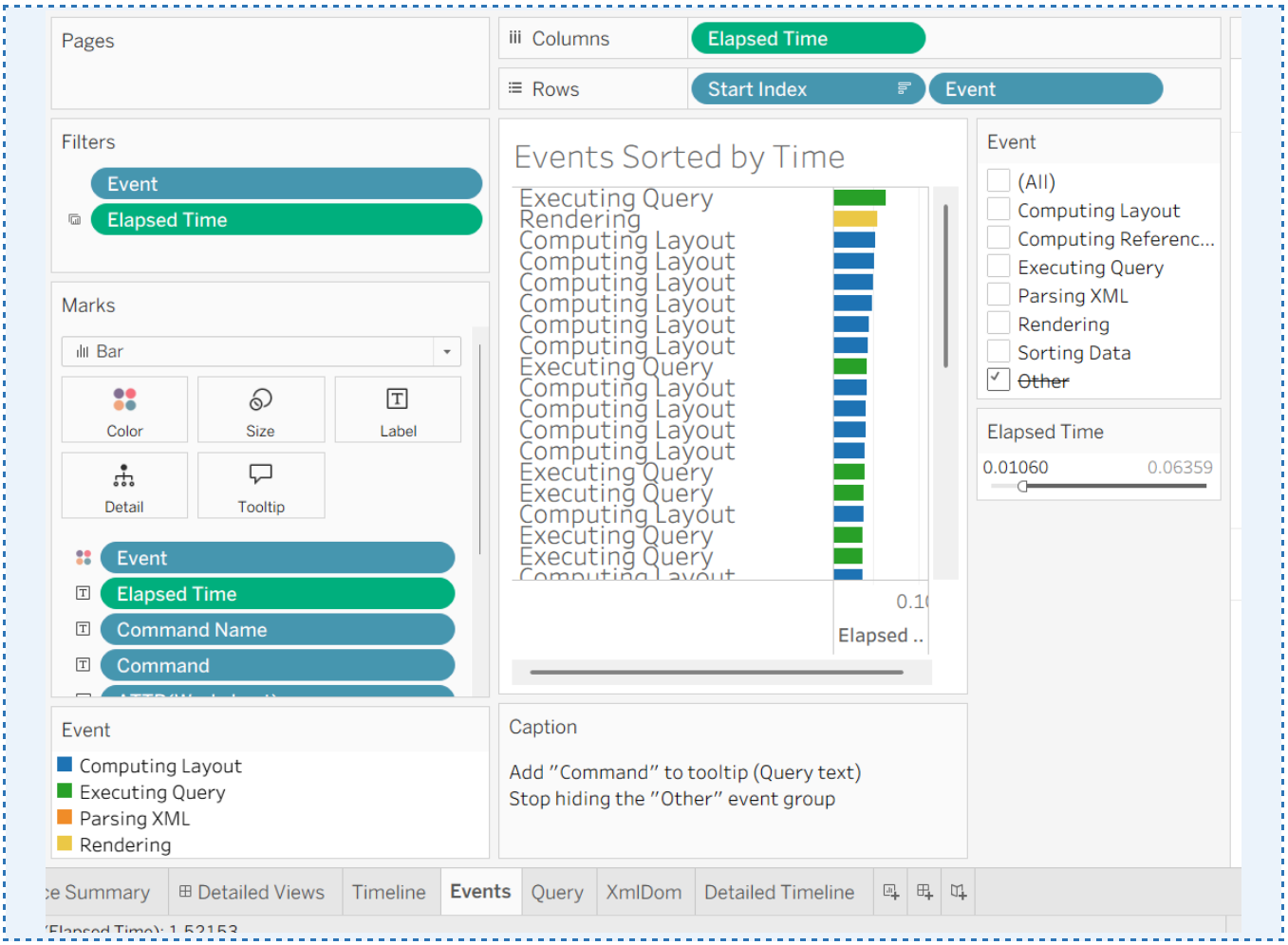
Key findings from performance recording:

- Query execution was the primary bottleneck, not rendering
- Switching from Live to Extract connection reduced average query time significantly
- LOD expressions were among the most compute-intensive calculations

8.2 Context Filters

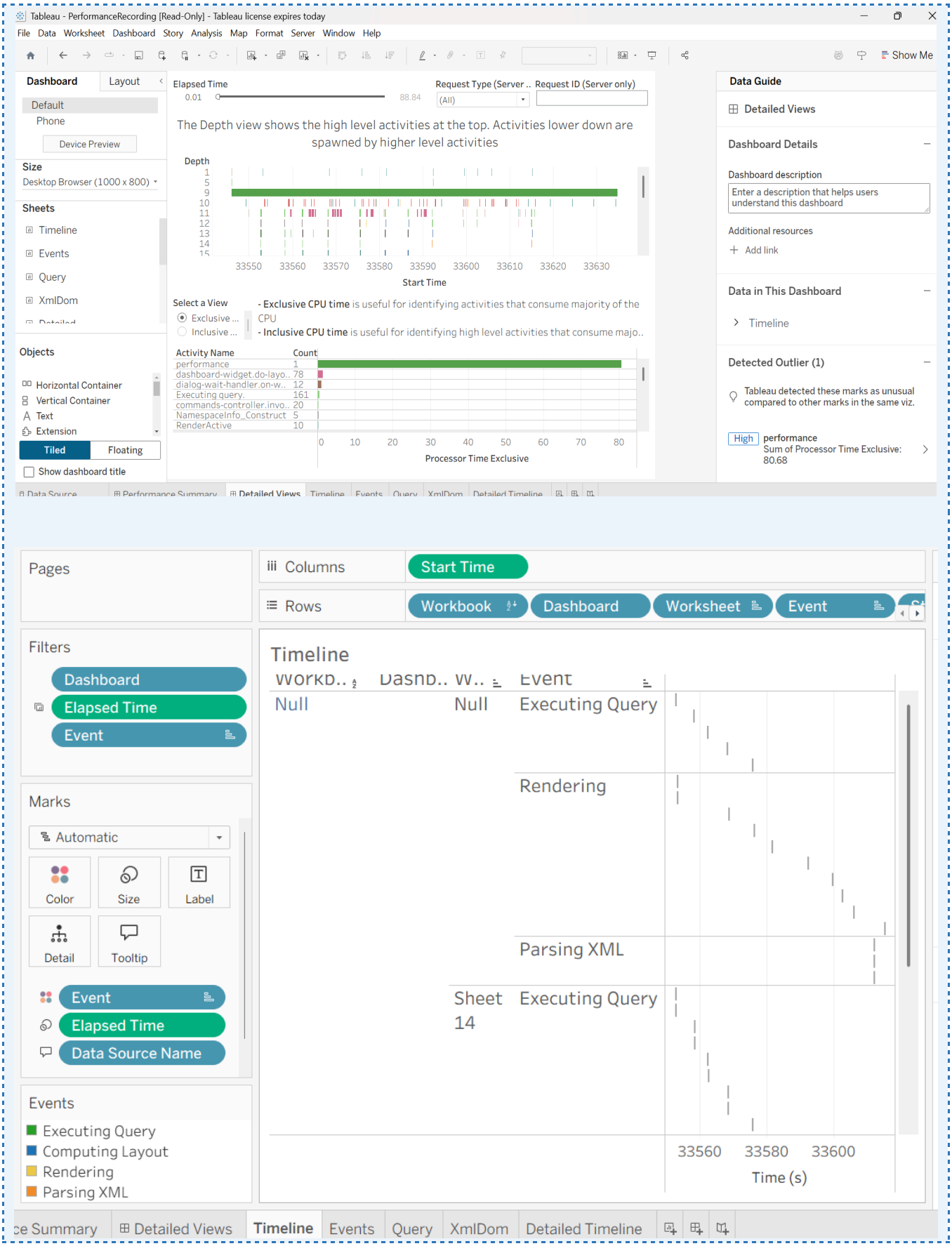
Context Filters were implemented to improve the accuracy and performance of dependent filters. A Context Filter runs first, reducing the dataset before other filters (including the Top N table calculation) are evaluated.

- Region = 'Europe' was added as a Context Filter on the Top N sheet
- The filter pill turned grey, confirming Context Filter status
- The Top N ranking now correctly ranks only within the filtered context



8.3 Data Source vs Extract Filters

Property	Data Source Filter	Extract Filter
Where Applied	On the live data connection	Applied during Extract (.hyper) creation
Performance	Query runs each time	Data pre-filtered — faster dashboards
Security	Restricts rows at connection	Reduces extract file size
Implementation	Data Source tab > Add Filter	Data Source tab > Extract > Edit > Add Filter
Used For	Row-level security	Performance + file size optimisation


Tableau - PerformanceRecording [Read-Only] - Tableau license expires today
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Show Me
Dashboard
Layout
Default
Phone
Device Preview
Size
Desktop Browser (1000 x 800)
Sheets
Timeline
Events
Query
XmlDom
Objects
Horizontal Container
Vertical Container
Text
Extension
Tiled
Floating
Show dashboard title
Elapsed Time
0.01
88.84
Request Type (Server ..
(All)
Request ID (Server only)
The Depth view shows the high level activities at the top. Activities lower down are spawned by higher level activities
Depth
Start Time
Select a View
Exclusive ...
Inclusive ...
- Exclusive CPU time is useful for identifying activities that consume majority of the CPU
- Inclusive CPU time is useful for identifying high level activities that consume majo..
Activity Name
Count
performance 1
dashboard-widget.do-layo.. 78
dialog-wait-handler.on-w.. 12
Executing query. 161
commands-controller.invo.. 20
NamespaceInfo_Construct 5
RenderActive 10
Processor Time Exclusive
Data Guide
Detailed Views
Dashboard Details
Dashboard description
Enter a description that helps users understand this dashboard
Additional resources
Add link
Data in This Dashboard
Timeline
Detected Outlier (1)
Tableau detected these marks as unusual compared to other marks in the same viz.
High
performance
Sum of Processor Time Exclusive: 80.68
Data Source
Performance Summary
Detailed Views
Timeline
Events
Query
XmlDom
Detailed Timeline
Pages
Columns
Start Time
Rows
Workbook
Dashboard
Worksheet
Event
Filters
Dashboard
Elapsed Time
Event
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Event
Elapsed Time
Data Source Name
Events
Executing Query
Computing Layout
Rendering
Parsing XML
Timeline
Null
Null
Executing Query
Rendering
Parsing XML
Sheet 14
Executing Query
33560
33580
33600
Time (s)
Summary
Detailed Views
Timeline
Events
Query
XmlDom
Detailed Timeline

9. Theoretical Questions

Q1. Explain Tableau Architecture

Tableau follows a three-tier client-server architecture:

Layer	Component	Role
Client Tier	Tableau Desktop / Reader / Mobile	User interface for authoring and viewing visualisations
Application Tier	Tableau Server / Tableau Online	Hosts workbooks, manages authentication, schedules refreshes
Data Tier	Data Sources / Hyper Engine	Stores and processes data via live connections or extracts

The VizQL engine sits at the core of Tableau Desktop, translating drag-and-drop actions into database queries and visual rendering instructions. The Data Server component manages data connections and extract caching on Tableau Server.

Q2. Explain Hyper Engine

Hyper is Tableau's proprietary in-memory analytical database engine, introduced in Tableau 10.5 to replace the older TDE (Tableau Data Extract) format. Key characteristics:

- Columnar Storage: data is stored column-by-column rather than row-by-row, enabling efficient compression and vectorised query execution.
- In-Memory Processing: queries are executed entirely in RAM for maximum speed, with intelligent spill-to-disk for datasets larger than available memory.
- Parallel Execution: Hyper uses multi-threaded query processing to exploit modern multi-core CPUs.
- Dictionary Encoding: repeating string values (e.g., country names) are stored as integer codes referencing a lookup table, reducing storage footprint significantly.
- Result: Hyper extracts are typically 5-10x faster to query than live connections and significantly smaller than equivalent CSV files.

Q3. Difference Between Data Blending and Joining

Aspect	Data Joining	Data Blending
Definition	Combines tables at the row level	Aggregates two sources separately, then links at view level
Data Source	Same data source	Different data sources (e.g., Excel + SQL)

Aspect	Data Joining	Data Blending
Join Type	Inner, Left, Right, Full Outer	Always a Left Join at the aggregate level
Performance	Faster — single query	Slower — two separate queries merged in Tableau
Granularity	Row-level combination	Aggregate-level combination
When to Use	Related tables in the same database	When data lives in different systems

In this project, Joining was used to combine the fact table with all four dimension tables within the same data source (all CSV files connected in the Tableau Data Source canvas).

Q4. Explain LOD Expressions

Level of Detail (LOD) expressions allow calculations at a different granularity than the current view, without requiring separate worksheets. Three types:

Type	Syntax	Behaviour	Responds to Filters?
FIXED	{ FIXED [dim] : AGG([measure]) }	Computes at specified dimension, ignores view	No (only Context Filters)
INCLUDE	{ INCLUDE [dim] : AGG([measure]) }	Adds a dimension to the current view level	Yes
EXCLUDE	{ EXCLUDE [dim] : AGG([measure]) }	Removes a dimension from the current view level	Yes

Performance impact: FIXED is the most efficient as it pre-computes at the data source level. INCLUDE and EXCLUDE are evaluated at query time and may slow down dashboards with many LOD expressions. Context Filters can be used to make FIXED expressions filter-aware.

Q5. Compare Tableau with Power BI and Qlik Sense

Criteria	Tableau	Power BI	Qlik Sense
Developer	Salesforce	Microsoft	Qlik Technologies
Pricing	Higher (Desktop + Server)	Low (free desktop, $10/user/mo)	Mid-range
Ease of Use	Moderate — drag-and-drop	Easy — Excel-like interface	Moderate — associative model
Data Model	Relational / Star Schema	DAX / Power Query	Associative QIX engine

Criteria	Tableau	Power BI	Qlik Sense
Visualisations	Best-in-class, highly customisable	Good, improving rapidly	Good, SmartSearch driven
ML Integration	R & Python scripts, Cluster	Azure ML, AutoML built-in	AutoML via Qlik AutoML
Collaboration	Tableau Server / Cloud	Power BI Service (Azure)	Qlik Cloud
Best For	Advanced analytics, data viz	Microsoft ecosystem users	Self-service, search analytics

For this project, Tableau was chosen for its superior visualisation capabilities, flexible LOD expressions, and robust cohort and statistical analysis features.

10. Business Insights

The following insights were derived from the Tableau dashboards and analytical calculations:

Revenue & Profitability

- Kosovo, Czech Republic, and Ukraine consistently ranked in the Top 3 countries by Total Revenue, collectively accounting for over 30% of European sales.
- Cosmetics and Office Supplies showed the highest profit margins, while Beverages and Cereal exhibited the lowest margins — indicating potential for pricing strategy revision.
- Online sales channel generated higher average order values than Offline, suggesting investment in digital sales infrastructure.

Customer Segmentation (RFM)

- Countries classified as Champions (RFM Score >= 300) should be prioritised for premium product launches and loyalty programmes.
- At Risk countries (Score 130-199) require targeted re-engagement campaigns to prevent churn.
- Geographic clustering revealed that Western European countries generally achieved higher RFM scores than Eastern European counterparts.

Anomaly Detection

- Z-Score analysis identified 3-4 months across the 2011-2013 period with statistically anomalous revenue spikes (Z > 2), likely driven by seasonal demand peaks.
- Months with Z-Score below -2 represent revenue troughs that correlate with Low-priority order periods — suggesting operational inefficiencies during slow seasons.

Forecasting

- Exponential smoothing forecast projects moderate revenue growth through 2018, with seasonal patterns consistent with historical data.
- The 95% confidence band widens significantly beyond 6 months, indicating increasing forecast uncertainty — decisions should be reviewed quarterly.

Appendix — Screenshot Index

The following table provides a complete reference for all screenshots included in this report:

#	Screenshot Description	Section
1	Star Schema — all 5 CSV tables in Excel	Section 2
2	Tableau Data Source connection screen	Section 2
3	Data Source canvas with all tables joined	Section 2
4	Extract vs Live toggle (Extract selected)	Section 3
5	Surrogate Key calculated field formula	Section 3
6	FIXED LOD formula in calculated field editor	Section 4.1
7	INCLUDE LOD formula	Section 4.1
8	EXCLUDE LOD formula	Section 4.1
9	Cohort Heatmap	Section 4.2
10	Survival / Retention Curve	Section 4.2
11	Revenue + Moving Average line chart	Section 4.3
12	Anomaly Detection dual-axis chart	Section 4.3
13	Forecast chart with exponential smoothing	Section 5
14	Forecast Options dialog	Section 5
15	K-Means Cluster scatter plot	Section 6.1
16	RFM Score bar chart	Section 6.3
17	RFM Cluster scatter plot	Section 6.3
18	KPI Card (Total Revenue)	Section 7.1
19	Top N Bar Chart with slider	Section 7.1
20	Parameter creation dialog	Section 7.1
21	Drill-down hierarchy with + button	Section 7.1
22	Full Executive Control Panel Dashboard	Section 7.1
23	Full Customer Intelligence Lab Dashboard	Section 7.2

#	Screenshot Description	Section
24	Discount Abuse bar chart	Section 7.3
25	Profit Leakage scatter plot	Section 7.3
26	Full Operational Risk Dashboard	Section 7.3
27	Performance Recording — Help menu	Section 8.1
28	Performance Recording results workbook	Section 8.1
29	Context Filter grey pill on Filters shelf	Section 8.2
30	Data Source Filter applied	Section 8.3
31	Extract Filter dialog	Section 8.3